MGMT6564

Global Supply Chain Management

Midterm notebook · BINUS

Ahmad Syamil · Sessions 1–14

SECONDBRAIN

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Course walkthrough

*This is your study path for Global Supply Chain Management (GSCM). Each topic follows the same logic: the **problem** it solves, the **big idea**, the **tools**, and the **result**. Then comes a plain-words explanation, a diagram, and the details to memorise. "S1–2" means Session 1–2; "CILO" means Course Intended Learning Outcome, the official learning goal a session serves. Every abbreviation is written out the first time it appears, and the glossary at the end lists them all.*

CHAPTER 1 · START HERE

How the whole course fits together

IN PLAIN WORDS

Supply chain management (SCM) is the course's central idea. Every session answers one question about the chain, in a natural order: first *what* a supply chain is, then how to *make* products in it, how to *run* it with processes and data, which *strategy* to choose, and finally how to keep that strategy *sustainable*, *safe* and workable *across borders*.

The lecture notes from Mr. Syamil add a "toolbox" of thinkers and models (Porter, Fisher, Lean, Six Sigma) that you can use in any essay. Forum 11 adds the forecasting numbers. If an exam question feels unfamiliar, place it on this map first: that tells you which concepts to bring in.

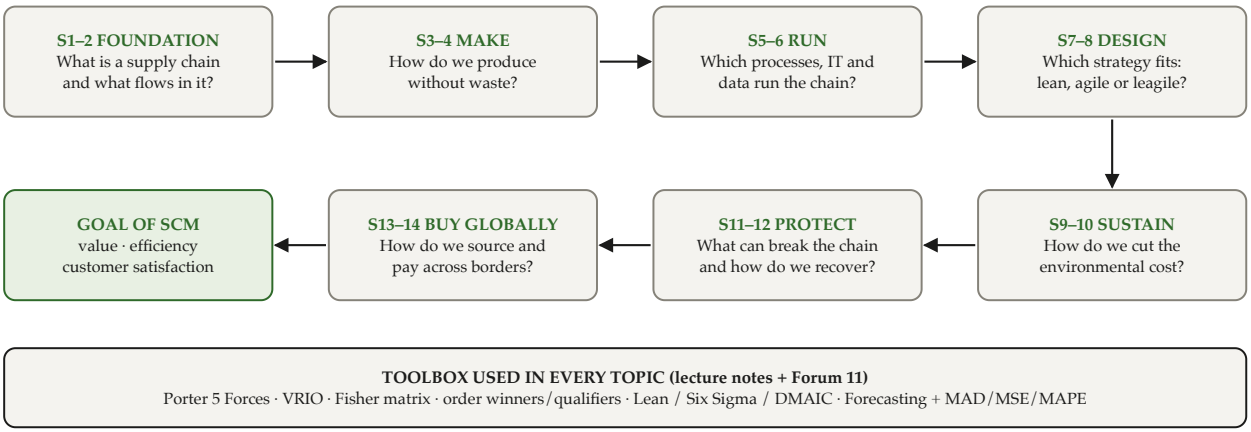


FIGURE 1.1 Read it like a story: define the chain → make things in it → run it with systems and data → choose a strategy → make that strategy green, safe and global. Toolbox terms: VRIO = Valuable, Rare, Inimitable, Organised; DMAIC = Define, Measure, Analyse, Improve, Control; MAD / MSE / MAPE = mean absolute deviation / mean squared error / mean absolute percentage error.

CHAPTER 2 · SESSION 1–2 · CILO 1: FUNDAMENTAL PRINCIPLES

What a supply chain is and what flows in it

PROBLEM	BIG IDEA	HOW (TOOLS)	RESULT
Firms no longer compete alone; products pass through many organisations before reaching the consumer.	Manage the whole network, not just your own company.	3 flows, 6 foundational premises, logistics as the material link.	Value created, efficiency raised, customers satisfied.

IN PLAIN WORDS

Think of a bottle of Aqua water. It involves a plastic maker, a bottling plant, a distributor (often a third-party logistics provider, or 3PL), a minimarket and you. That whole network is the **supply chain**. **Logistics** is the part that physically moves and stores the bottles, aiming to deliver the "7 Rights": the right product, in the right way, quantity and quality, at the right place and time, to the right customer, at the right cost.

Supply chain management is bigger. It manages the *relationships* between all those firms and three flows: **materials** (moving downstream to the customer), **information** (orders, forecasts and point-of-sale (POS) checkout data, moving both ways) and **resources** such as money, people and equipment (payments move upstream). So logistics is one component inside SCM, a point the slides stress.

Logistics and supply chain management (LSCM) grew important because of six developments: freight became less transport-intensive, product prices fell (squeezing margins), transport was deregulated, productivity rose, firms focused on cutting inventory, and company structures changed (more outsourcing). The six **foundational premises (FP)** describe how a chain behaves: it is a network that no single firm fully controls, and each firm sees only part of it.

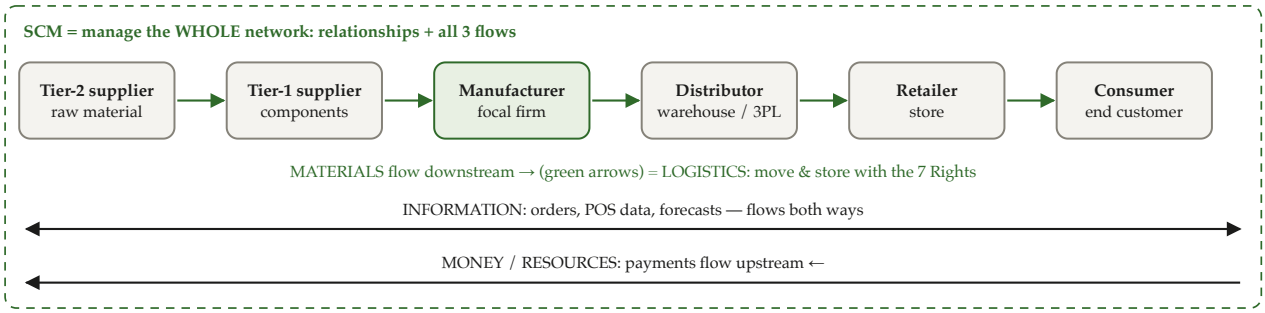


FIGURE 2.1 Logistics is only the green links (moving and storing goods). SCM is the dashed box: every node, every relationship and all three flows. That is why "logistics is a component of SCM".

Must-know definitions

- **Supply chain**: the network of organisations linked upstream and downstream that produce value for the ultimate consumer.
- **SCM purpose**: create value, enhance efficiency, satisfy customers.
- **Value chain vs supply chain**: the value chain (activities and competencies that add value) *creates* the flows; the supply chain *operationalises* them.
- **Sector names**: farm to fork (food), sketch to store (clothing), dust to rust (cars and machinery), teeth to tail (military).
- **Applications**: cost savings plus service add-ons (after-sales, delivery). IKEA flat-pack; hospital triage that matches patients to the right care stream early.

Six developments behind LSCM

- Reduced transport intensity of freight
- Falling product prices
- Deregulation of transport
- Productivity improvements
- Emphasis on inventory reduction
- Changes in company structure

Six foundational premises: read them as one argument

1 Network	2 Complex	3 Relative	4 Two layers
FP1: nodes (firms) and links (flows).	FP2: a complex adaptive system; tension between	FP3: a chain is defined from one	FP4: physical chain + support chain (banks,

	control and emergence.	product and one agent.	IT, logistics providers).

5 Limited view
FP5: bounded by what the focal firm can see.

6 Fading view
FP6: the view fades with physical distance, cultural distance and closeness centrality.

Exam tip: For "distinguish logistics and SCM", describe the diagram: logistics = the material links (7 Rights); SCM = the whole box, including relationships and all three flows.

Link to next topic → Once you know what the chain is, the next question is how to make products in it efficiently → lean.

CHECK YOURSELF

Answer from memory, then check below.

1. Foundational premise FP5 says the supply chain is…

A. A network of nodes and links
B. Relative to a product and agent
C. Bounded by the focal agent's visible horizon
D. A physical chain plus a support chain

2. 'Sketch to store' describes the supply chain of…

A. Food
B. Clothing
C. Cars and machinery
D. Military

3. RECALL 6 developments behind logistics and SCM

Answers 1. C **2.** B **3.** Reduced transport intensity of freight, falling product prices, deregulation of transport, productivity improvements, emphasis on inventory reduction, changes in company structure.

CHAPTER 3 · SESSION 3–4 · CILO 1

Making things without waste: lean

PROBLEM
Mass production gave low cost but huge stocks, defects and slow change.

→

BIG IDEA
Produce only what the customer pulls; anything that adds no value is waste.

→

HOW (TOOLS)
5 lean principles, 7+1 wastes, just-in-time, level production, quality tools.

→

RESULT
Lower cost and inventory, better quality, faster response.

IN PLAIN WORDS

Production evolved from **craft** (custom, one by one) to **mass production** (Henry Ford: standard products, economies of scale), to **flexible manufacturing**, to **lean** (Toyota), and now to Industry 4.0 (automation, digital, sustainability). Each step tried to fix the weakness of the one before.

Mass production is a **push** system: you build to a forecast and hold the result as stock, called make-to-stock (MTS). If the forecast is wrong, you get piles of unsold goods. Lean flips this to **pull**: production starts only when a real order arrives, called make-to-order (MTO), with materials arriving just-in-time (JIT) and production kept level (even, with no idle time and no surges).

Lean sees anything the customer would not pay for as **waste** (Japanese *muda*). Removing waste lowers cost and speeds flow. Because quality and price are now expected by every buyer, the slides say "order winners have become order qualifiers": firms now compete through their whole supply chain. Quality approaches here are total quality management (TQM), *kaizen* (Japanese for continuous small improvement) and Lean Six Sigma.

History in one line

1. **Craft** — custom, low volume
2. **Mass** — standard, economies of scale
3. **Flexible** — variety with automation
4. **Lean** — waste out, continuous improvement
5. **Industry 4.0** — digital, automated, sustainable

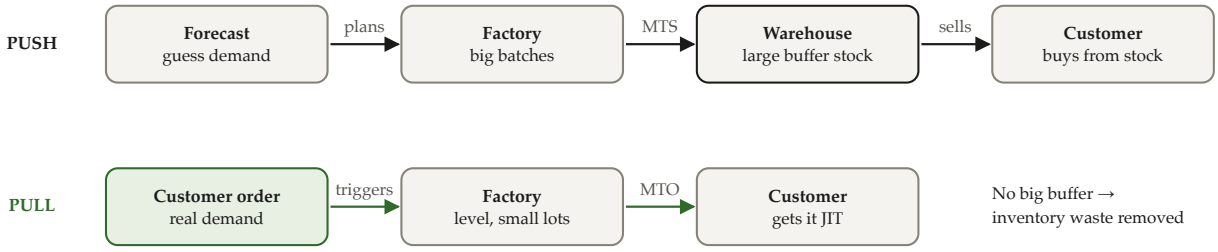


FIGURE 3.1 The whole lean logic in one picture: push builds stock on a guess (risk of overproduction and inventory waste); pull waits for the real order, so the buffer, and the waste in it, disappears.

5 lean principles (in order)

1. **Value** — define it from the customer view
2. **Value stream** — map every step that creates it
3. **Flow** — make steps flow; remove waiting
4. **Pull** — produce on real demand
5. **Perfection** — find root causes, keep improving

8 wastes: mnemonic TIM WOODS

WASTE	SUPPLY CHAIN EXAMPLE
Transport	double handling between warehouses
Inventory	excess safety stock
Motion	pickers walking long routes
Waiting	trucks queuing at docks
Overproduction	making before orders arrive
Over-processing	excess packaging, extra checks
Defects	damaged goods, returns
Skills (8th)	underutilised people and resources

- **4 Vs of process design**: Volume, Variety, Variation in demand, Visibility of the customer. High volume and low variety suit standard, automated processes (a bottling line); low volume and high variety need flexible processes (a tailor).
- **Lean consumption** applies lean to the customer: solve their problem completely, don't waste their time, provide exactly what they want, where and when they want it, and keep reducing their hassle.
- **Servitisation**: selling services together with products. Anything-as-a-service (XaaS), e.g. mobility as a service (MaaS, like Gojek instead of owning a car) and supply chain as a service (SCaaS).
- **Quality tools**: Pareto analysis (fix the few causes behind most problems) and importance-performance analysis (improve what customers rate important but

you perform poorly on). See the Improvement section.

Link to next topic → Lean needs accurate, shared information to pull on real demand → systems, analytics and process management.

CHECK YOURSELF

Answer from memory, then check below.

1. Lean production shifts a firm from…

A. Pull to push
B. Make-to-order (MTO) to make-to-stock (MTS)
C. Push to pull and MTS to MTO
D. Level production to batch production

2. Excess finished goods built 'just in case' is which waste?

A. Waiting
B. Overproduction
C. Motion
D. Defects

3. RECALL **Evolution of production strategies**

Answers 1. C **2.** B **3.** Craft (custom, low volume) → mass production (standardisation, economies of scale) → flexible manufacturing → lean (waste elimination, continuous improvement) → digital: automation, Industry 4.0, sustainability.

CHAPTER 4 · SESSION 5–6 · CILO 1

Running the chain: processes, IT and analytics

PROBLEM	BIG IDEA	HOW (TOOLS)	RESULT
Many partners, lots of data, but poor visibility and slow decisions (e.g. aerospace before AirSupply).	Manage processes systematically and decide with data.	Process management, information systems, problem-solving cycles, analytics, simulation, optimisation.	Visibility, collaboration, lower inventory and cost, faster response.

IN PLAIN WORDS

AirSupply is the opening case: one online platform where aerospace makers (Airbus and its suppliers) share forecasts, track orders and run vendor-managed inventory (VMI), where the supplier watches and refills the customer's stock. It works as early warning, and everyone sees problems before they become shortages.

Three ideas make this work. **Business process management (BPM)** treats each process (procurement, order fulfilment) as something to model, analyse, redesign, execute and monitor in a loop. **Management information systems (MIS)** collect the data: enterprise resource planning (ERP) software, supplier portals, internet of things (IoT) sensors, cloud platforms. **Business analytics** turns that data into decisions, measured with key performance indicators (KPIs).

When something goes wrong (bottlenecks, stockouts, delays), you use problem-solving cycles. Plan-Do-Check-Act (PDCA) runs quick small experiments. Six Sigma's Define-Measure-Analyse-Improve-Control (DMAIC) is more data-heavy. Root-cause tools include the 5 Whys and the fishbone diagram. For big design questions, **simulation** tests "what if" safely, and **optimisation** (linear or integer programming) finds the best answer under constraints.

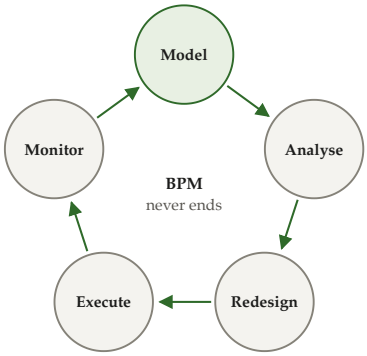


FIGURE 4.1 BPM: map the process, find the problem, redesign, run it, measure, repeat.

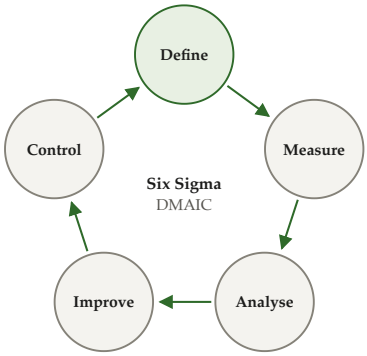


FIGURE 4.2 DMAIC: data-heavy, finds the root cause before fixing.

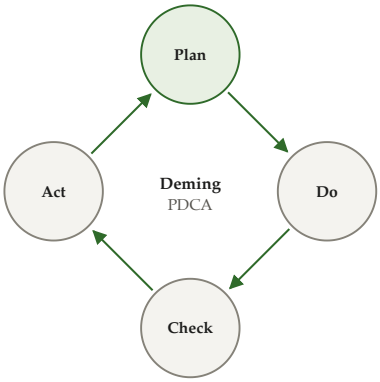


FIGURE 4.3 PDCA: quick, repeated small experiments (kaizen).

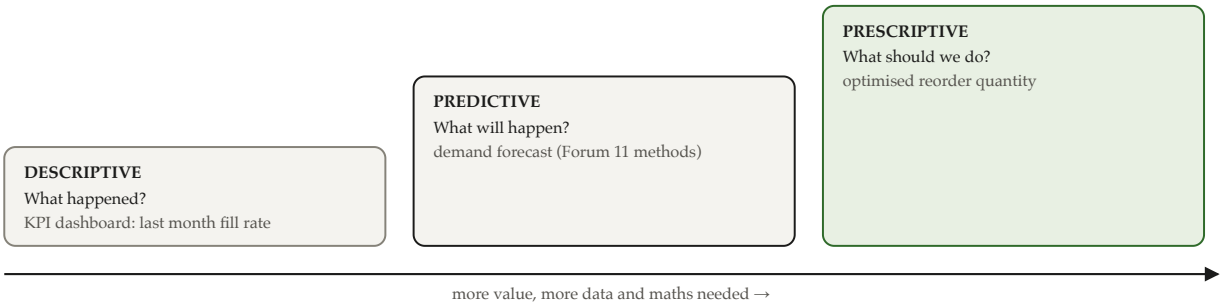


FIGURE 4.4 Each step builds on the one below: you cannot predict without describing the past, and you cannot prescribe an action without a prediction. Forecasting (Forum 11) is the predictive step.

Systems and models

- **MIS types**: ERP, SCM execution systems, supplier portals → real-time visibility and exception management.
- **Technologies**: IoT sensors for inventory tracking, cloud platforms, artificial intelligence and machine learning (AI/ML) for forecasting.
- **Simulation** (discrete-event, agent-based; software AnyLogic, Arena, Simul8): capacity planning, risk evaluation, layout tests.
- **Optimisation**: network design (where to put warehouses), inventory (holding vs stockout cost), scheduling.
- **Emerging**: digital twins (live virtual copy of the chain), blockchain (secure shared records), AI, sustainability analytics.
- **Research methods**: quantitative (statistics, optimisation), qualitative (interviews, case studies), mixed methods.

4 challenges

- **Data quality & integration**: incomplete or inconsistent data; partners' IT systems don't talk → silos.
- **Modelling complexity**: global, dynamic, uncertain chains are hard to model; needs computing power and expertise.
- **Resistance to change**: fear of job loss, habit, low trust, little training.
- **Robustness vs agility**: robustness needs spare capacity/stock (cost); agility needs flexible processes (complex). Models must balance cost, service, risk and flexibility.

> **Exam tip:** Name the analytics type and give one supply chain example each. Link predictive to forecasting and prescriptive to optimisation.

Link to next topic → With processes and data in place, the firm must choose HOW to compete → supply chain strategy.

CHECK YOURSELF

Answer from memory, then check below.

1. **In the AirSupply case, the supplier monitors and refills the customer's stock. This is called…**
   - A. Enterprise resource planning (ERP)
   - B. Vendor-managed inventory (VMI)
   - C. Business process management (BPM)
   - D. Cross-docking

2. **Recommending how much stock to reorder next week is which type of analytics?**
   - A. Descriptive
   - B. Diagnostic
   - C. Predictive
   - D. Prescriptive

3. RECALL **Management information systems (MIS) in SCM**

Answers 1. B **2.** D **3.** Computerised systems that collect and process data for decisions: enterprise resource planning (ERP), SCM execution systems, supplier portals. They give real-time visibility and exception management.

CHAPTER 5 · SESSION 7–8 · CILO 1

Choosing a strategy: outsourcing, lean vs agile, leagile

PROBLEM
One chain cannot be both cheapest and fastest for every product.

→

BIG IDEA
Match the chain to the product and decide what to do yourself vs buy in.

→

HOW (TOOLS)
Outsourcing + service level agreements, supplier selection, offshoring, postponement, leagile.

→

RESULT
Cost efficiency where demand is stable, agility where it is not.

IN PLAIN WORDS

Strategy flows top-down: corporate strategy → business strategy → functional strategies such as logistics. The first big decision is **make or buy**. **Outsourcing** transfers a process to another company, often a third-party logistics provider (3PL) that runs transport or warehousing. A firm that outsources almost everything is a **virtual organisation**. To keep control, the buyer writes a **service level agreement (SLA)**: a contract that sets measurable standards (on-time delivery %, inventory accuracy, response speed) and what happens if they are missed.

Relationships with suppliers mature over four stages, and suppliers are organised in **tiers**: tier 1 supplies you directly, tier 2 supplies tier 1. A **module** is a complete sub-assembly, such as a car dashboard, supplied by a lower tier. **Offshoring** is different from outsourcing: it moves work to *another country*, usually for lower cost, and brings culture, time-zone and regulation challenges.

The core trade-off is **lean vs agile**. Lean is cheapest when demand is stable; agile (responsive, flexible) is needed when demand is unpredictable. **Leagile** combines them: be lean upstream, where you make a generic base product, and agile downstream, where you customise it to real orders. The switch point is the **decoupling point**. **Postponement** deliberately delays customisation as far downstream as possible. Examples: paint mixed in-store to the customer's colour, or a laptop configured after the order. Postponement is what enables **mass customisation**.

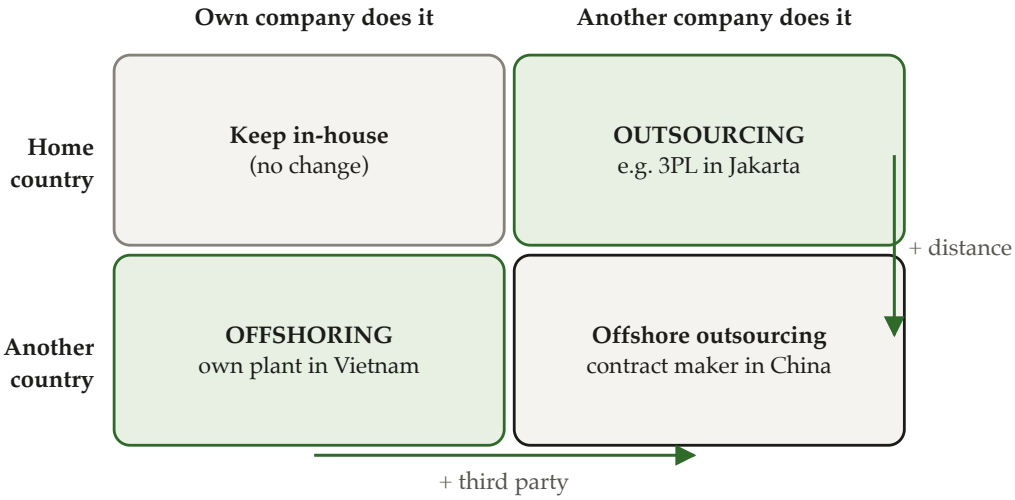


FIGURE 5.1 Outsourcing answers "who?" (a third party). Offshoring answers "where?" (another country). They can combine, which is the riskiest and usually cheapest cell.

What an SLA contains

- Roll-out and duration of the service
- Scope of services
- Areas of responsibility and legal requirements
- Performance metrics
- Termination / exit arrangements

Outsourcer–supplier relationship stages

1. **Master-servant** transactional; buyer dictates, one-way
2. **Consultative** supplier input asked; buyer still leads
3. **Peer-to-peer** equals; trust, joint decisions, shared risk & reward
4. **Competitive** several suppliers compete to drive improvement

Supplier selection (order qualifiers): capability, delivery reliability, quality certifications, conformance to specifications, delivery lead time, financial capability, performance track record, price or cost reduction, senior management attitude, responsiveness to demand uncertainty, ethical requirements.

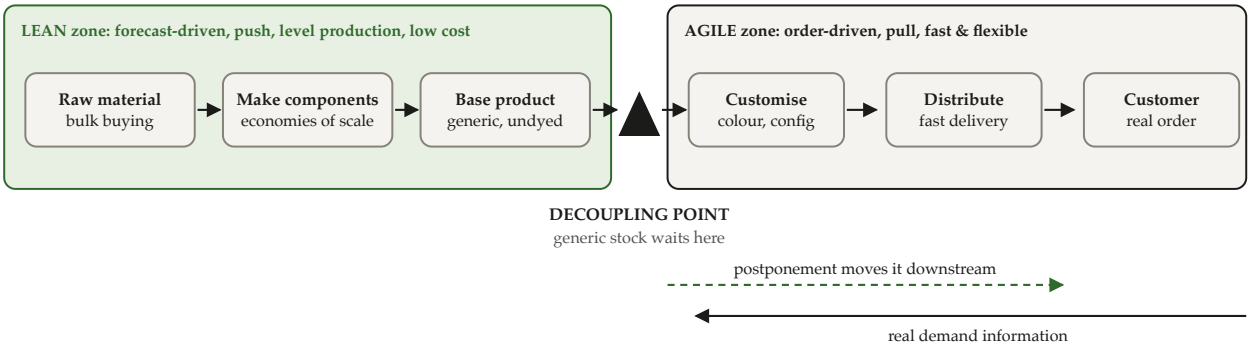


FIGURE 5.2 Left of the triangle you can be efficient because demand for the generic product is stable. Right of it you must be agile because customers want specific variants. Postponement = keep products generic for longer, so less risky stock is held.

- **Responsiveness** = meeting customer demand quickly. **Agility** = adapting to market changes and disruptions. Modern strategy balances both against cost.
- **Integration & collaboration**: real-time information sharing, joint planning, operational coordination → better visibility, less inefficiency, faster response.

> **Exam tip:** In any leagile or postponement answer, draw the triangle: lean on the left, agile on the right, decoupling point = generic stock, postponement pushes it right. Fisher's matrix (Strategy toolbox) explains *why*.

Link to next topic → A good strategy must also be sustainable → environmental cost and ESG.

CHECK YOURSELF

Answer from memory, then check below.

1. **Which is NOT listed as part of a service level agreement (SLA)?**
   - A. Scope of services
   - B. Performance metrics
   - C. Termination/exit arrangements
   - D. The supplier's marketing plan

2. **Buyer and supplier interact as equals, with joint decisions and shared risk and reward. Which stage?**
   - A. Master-servant
   - B. Consultative
   - C. Peer-to-peer
   - D. Competitive

3. RECALL 5 contents of a service level agreement (SLA)

Answers 1. D **2.** C **3.** Roll-out and duration; scope of services; areas of responsibility and legal requirements; performance metrics; termination/exit arrangements.

CHAPTER 6 · SESSION 9–10 · CILO 2: ANALYSE AND EVALUATE

Sustainability: cutting the environmental cost

PROBLEM	BIG IDEA	HOW (TOOLS)	RESULT
Moving freight creates emissions and waste; regulators, investors and customers now demand action.	Measure the footprint across the whole chain, then redesign to reduce it and close the loop.	ESG + CSR, Scope 1/2/3, carbon footprint formula, reverse logistics, 3Rs.	Lower emissions and risk, better reputation, compliance.

IN PLAIN WORDS

Sustainable logistics reduces the environmental and other "disbenefits" of moving freight, so today's decisions don't harm future generations. Two frameworks guide companies. **Corporate social responsibility (CSR)** is the voluntary ethical commitment: the values and vision. **Environmental, social and governance (ESG)** criteria are the measurable targets that put CSR into practice and let investors check it. In short, CSR is the "why" and ESG the "how much".

To manage emissions you first measure them. Greenhouse gases (GHG) are reported as carbon dioxide equivalent (CO_2eq), which converts methane and nitrous oxide into the same warming effect as CO_2. They are split into three **scopes** by asking "whose activity is it?" Scope 1 is your own fuel, Scope 2 is electricity you buy, and Scope 3 is everyone else in your chain. Scope 3 is the hardest to measure but often the largest share.

Finally, **reverse logistics** moves products back upstream for return, reuse, repair, remanufacture or recycling. It is the backbone of the **circular economy** and its 3Rs: Reduce, Reuse, Recycle.

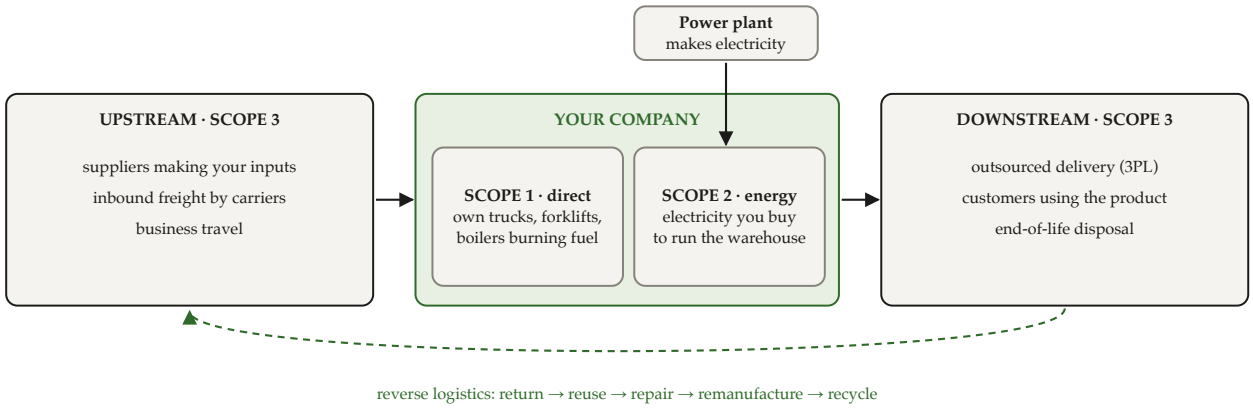


FIGURE 6.1 Ask "whose chimney?": your own fuel = Scope 1, your bought electricity = Scope 2, anyone else's activity in your chain = Scope 3. The dashed loop is reverse logistics, the basis of the circular economy.

ESG in detail

- **Environmental**: carbon emissions, resource use, waste, biodiversity.
- **Social**: fair labour, community engagement, human rights.
- **Governance**: ethical leadership, legal compliance, stakeholder accountability.

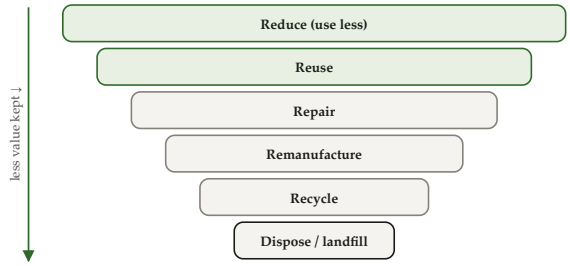


FIGURE 6.2 Higher = keeps more value and creates less waste. The 3Rs (reduce, reuse, recycle) sit in this order on purpose.

Carbon footprint = Fuel used × Emission factor
1,000 L diesel × 2.68 kg CO_2eq/L ≈ 2,680 kg CO_2eq

About 95% of GHG emissions are CO_2, mostly from burning fossil fuels.

How to improve: redesign supply chains (remove unnecessary freight movements), use less harmful energy, use scale (move freight in larger single loads, which cuts cost and emissions per unit), improve efficiency, and develop more fuel-efficient engines and alternative fuels.

> **Exam tip:** The same truck can be Scope 1 for the carrier and Scope 3 for the retailer that hired it. Always say whose report you mean.

Link to next topic → Besides environmental cost, the chain faces disruptions → complexity and risk.

CHECK YOURSELF

Answer from memory, then check below.

1. **Which statement about corporate social responsibility (CSR) and environmental, social and governance (ESG) criteria is correct?**
   - A. They are identical
   - B. CSR sets the ethical vision; ESG provides measurable targets
   - C. ESG is voluntary; CSR is law
   - D. CSR covers only governance
2. **Electricity bought to power a warehouse is…**
   - A. Scope 1
   - B. Scope 2
   - C. Scope 3
   - D. Carbon neutral
3. RECALL **Carbon footprint formula**

Answers 1. B **2.** B **3.** Fuel used × emission factor for that fuel, in kg CO_2eq (carbon dioxide equivalent, which includes methane and nitrous oxide).

CHAPTER 7 · SESSION 11–12 · CILO 2

Complexity and risk: what breaks the chain

PROBLEM	BIG IDEA	HOW (TOOLS)	RESULT
Long, complex global chains amplify small problems and break under shocks.	Reduce complexity, see further, build in flexibility and back-ups.	Bullwhip fixes, risk management tools, security standards, diversification, VUCA mindset.	A robust and resilient chain.

IN PLAIN WORDS

A supply chain sits inside bigger systems (geopolitics, ecosystems, politics, regulation), so shocks from outside hit it. The **bullwhip effect** shows how fragile a chain is even without shocks. Consumers buy steadily, but each level reacts to the orders of the level below. It adds a safety margin, orders in batches, stocks up on promotions and over-orders when it fears a shortage. So small consumer changes grow into wild swings at the factory.

Complexity makes this worse: too many nodes, steps, products, components, customer options, suppliers, organisational layers or data streams. Three terms describe how a chain copes. **Vulnerability** is how likely it is to be damaged or disrupted. **Robustness** is handling normal demand fluctuations. **Resilience** is returning to the original or desired state after a disturbance.

Supply chain risk management (SCRM) uses visibility, flexibility, postponement, multiple sourcing, buffers and joint planning. The world is also **VUCA**: volatile, uncertain, complex and ambiguous. The suggested response is vision, understanding, clarity and agility.

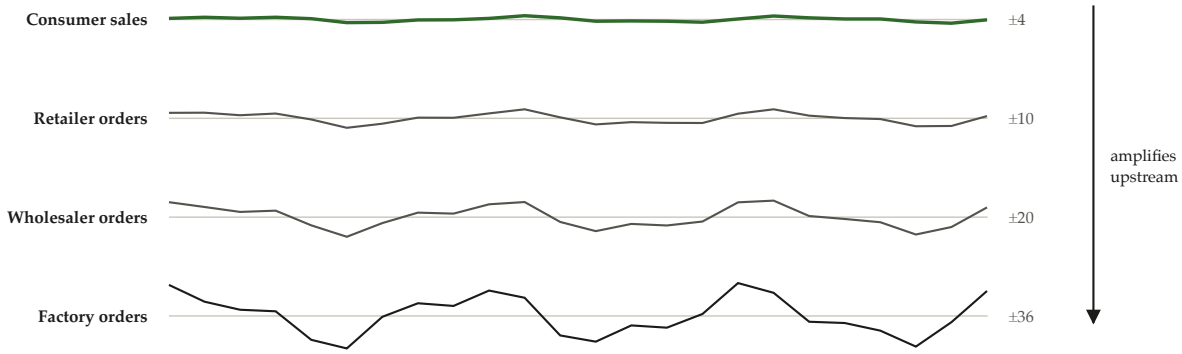


FIGURE 7.1 Same weeks, same scale, four levels of the chain. Consumers barely change what they buy, but each level over-reacts (forecast updating, batching, promotions, shortage gaming), so the factory sees huge swings. Illustrative data.

Bullwhip: causes → fixes

CAUSE	FIX
Demand forecast updating at each level	share POS data along the chain
Order batching	smaller, more frequent orders
Price fluctuations (promotions)	everyday low prices
Rationing and shortage gaming	allocate on past sales; VMI

Causes from Hau Lee's classic article (lecture notes).

8 complexity types

Network (too many nodes and links), process (too many steps), range (too many products), product (too many unique parts), customer (too many service options), supplier (too many suppliers), organisational (too many levels and silos), information (too much, often inaccurate, data).

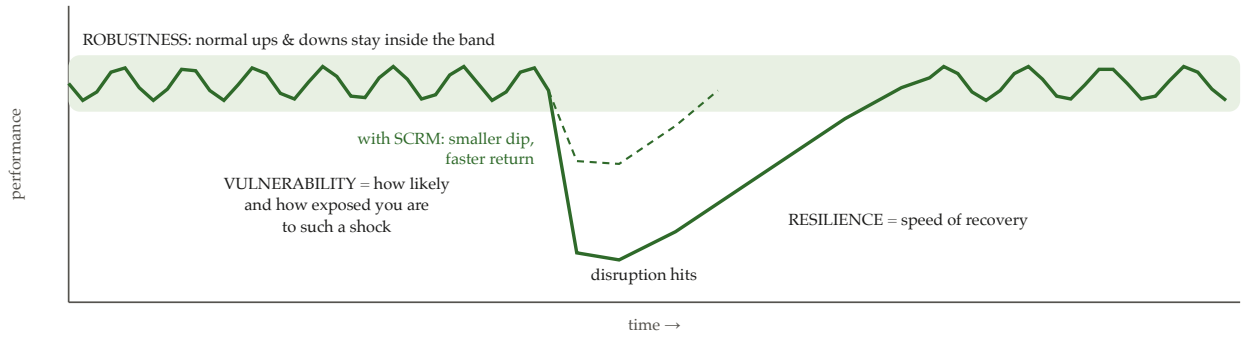


FIGURE 7.2 Three different words, three different parts of the same curve. SCRM tools (visibility, safety stock, multiple sourcing, postponement) make the dip shallower (dashed) and the recovery faster.

SCRM toolkit

- Visibility through information sharing with suppliers and clients
- Flexible production
- Postponement
- Multiple sourcing and flexible vendor contracts
- Redundancy: safety stock (e.g. through VMI)
- Joint planning and coordination

Risk types: systemic, market, operational, credit & liquidity.
Unknowns: known unknowns, knowable unknowns, unknowable unknowns, creeping crises.
Vulnerability questions: what disrupted us before? known weaknesses? near misses? effect of losing a key material, site, supplier or customer?

VUCA → response

Volatility	→ Vision
Uncertainty	→ Understanding
Complexity	→ Clarity
Ambiguity	→ Agility

Security technology: access control, tracking and tracing, detection systems.
Initiatives: the International Maritime Organization's (IMO) International Ship and Port Facility Security (ISPS) code; the US Customs-Trade Partnership Against Terrorism (C-TPAT), Container Security Initiative (CSI) and Air Cargo Advance Screening (ACAS); the EU Authorised Economic Operator (AEO); and the International Organization for Standardization's ISO 28000 for supply chain security.

Diversification: multishoring, multisourcing, varied logistics infrastructure, several transport modes in parallel.

Link to next topic → Global risk is highest where you buy across borders → international purchasing.

CHECK YOURSELF

Answer from memory, then check below.

1. **Retailer orders swing more than consumer sales, and factory orders swing even more. This is the…**
   A. Bullwhip effect
   B. Postponement principle
   C. Kaizen effect
   D. Leagile effect

2. **'Too wide a range of products offered' is which type of complexity?**

A. Product
B. Range
C. Customer
D. Network

3. RECALL **Transport security initiatives**

Answers 1. A **2.** B **3.** IMO's ISPS code (International Maritime Organization, International Ship and Port Facility Security); US C-TPAT (Customs-Trade Partnership Against Terrorism), CSI (Container Security Initiative), ACAS (Air Cargo Advance Screening); EU AEO (Authorised Economic Operator); ISO 28000.

CHAPTER 8 · SESSION 13–14 · CILO 2–3: APPLY TO IMPROVE EFFICIENCY

Buying globally: international purchasing

PROBLEM
Buying across borders adds distance, currencies, laws, documents and payment risk.

→

BIG IDEA
Treat purchasing as a planned process integrated with logistics and finance.

→

HOW (TOOLS)
8-step process, logistics providers, trade terms, payment methods, duty-free storage.

→

RESULT
Reliable, lower-cost supply with controlled risk.

IN PLAIN WORDS

International purchasing is obtaining goods or services from a market that requires crossing international borders. It needs many skills at once: logistics, negotiation, language and culture, technical specifications, supplier audits, standards and trade finance (paying in the buyer's, seller's or a third currency).

The buyer follows a planned process (below). Key partners are 3PLs and **fourth-party logistics providers (4PL)**, which integrate and manage the whole chain, often coordinating several 3PLs. Contracts specify **Incoterms** (International Commercial Terms: standard rules on who pays for and carries the risk of transport at each point). Letter-of-credit documents follow the **UCP 600** rules (Uniform Customs and Practice for Documentary Credits, publication no. 600).

Goods are cleared through customs at seaports or **inland container depots (ICD)**, inland "dry ports". Importers can store goods in a **free trade zone (FTZ)** or bonded warehouse, paying import duty only when goods are withdrawn. The last piece is money: choosing a payment method, managing foreign exchange (FX) risk, and using trade credit and insurance.

The 8-step process

1. **Identify needs** — specification, volume, quality, standards
2. **Find suppliers** — directories, associations, exhibitions; choose 3PL/4PL
3. **Plan negotiation** — price, Incoterms, payment, insurance, documents, delivery
4. **Activate contract** — dates, quantities, issuing bank, UCP 600 documents
5. **Manage logistics** — collection, customs at ICD/seaport, transshipment, FTZ
6. **Track cargo** — online tracking by 3PL/4PL
7. **Receive & evaluate** — inspect quality, pay, clear customs
8. **After-care** — benchmark performance, continuous improvement



FIGURE 8.1 Risk moves between the two parties. New relationship or risky country → L/C. Long, trusted relationship → open account (cheapest, but the seller carries the risk).

10 strategic areas

(1) Positioning the firm: in-house vs outsourced; (2) product design strategy; (3) supply chain environment in developed vs less-developed countries; (4) marketing–logistics collaboration; (5) trade-offs: near the market vs distant sourcing with longer lead times but lower cost (Switzerland precision engineering, Italy design, Germany heavy engineering, China low-cost mass production); (6) project management and value engineering; (7) structural planning (containerisation, warehousing, FTZ/ICD, finance, insurance); (8) planning and control; (9) human resource management (HRM); (10) location of factories and distribution centres (DC).

Financing the chain

- **Letter of credit (L/C)**: the buyer's bank guarantees payment once correct documents are presented. The most common instrument.
- **Documentary collection**: banks release shipping documents (needed to claim the goods) only against payment or acceptance.
- **Open account**: the buyer pays after receiving goods; highest risk for the supplier.
- Plus FX management, trade credit and supplier financing, insurance and guarantees, and banks' oversight.

Contract negotiation covers: commercial, technical, legal and transport aspects; standards; sourcing; tendering; supplier audit; costing; quality and dispute procedures; security.

CHECK YOURSELF

Answer from memory, then check below.

1. **Storing imported goods so that duty is paid only when they are withdrawn uses a…**
   - A. Cross-dock
   - B. Free trade zone (FTZ) or bonded warehouse
   - C. Hub-and-spoke network
   - D. Inland container depot only
2. **Which payment method carries the highest risk for the supplier (exporter)?**
   - A. Letter of credit (L/C)
   - B. Documentary collection
   - C. Open account
   - D. All equal
3. RECALL **3 international payment methods**

Answers 1. B **2.** C **3.** Letter of credit (L/C: buyer's bank guarantees payment), documentary collection (export documents as payment security), open account (pay after receipt; highest risk for the supplier).

CHAPTER 9 · LECTURE NOTES · USED ACROSS ALL SESSIONS

Strategy toolbox (Mr. Syamil's notes)

PROBLEM
How does a firm decide where and how to compete?

→

BIG IDEA
Look outside (industry), look inside (resources), then position operations to win orders.

→

HOW (TOOLS)
Porter, resource-based view, order winners/qualifiers, competitive priorities, Fisher, balanced scorecard.

→

RESULT
A strategy that operations can actually deliver.

IN PLAIN WORDS

Operations management (OM) is the design, operation and improvement of the systems that transform inputs into outputs of greater value. The transformation can be physical (factory), locational (transport), exchange (retail), physiological (hospital), psychological (entertainment) or informational (telecom). Productivity measures how well you transform: output divided by input.

Strategy starts with two views. The **outside view** is Michael Porter's Five Forces, which shows how attractive an industry is. The **inside view** is the **resource-based view (RBV)**: your advantage comes from resources that pass the **VRIO** test (Valuable, Rare, Inimitable, Organised to capture value). Next, decide what customers need just to consider you (**order qualifiers**) and what makes them choose you (**order winners**). Then position on cost, speed, quality or flexibility.

Marshall Fisher adds the supply chain link. Products with stable demand (functional) need an efficient chain; products with unpredictable demand (innovative) need a responsive chain. Strategy is turned into action through **Hoshin Kanri** (Japanese for "policy management", also called policy deployment) and tracked with the **balanced scorecard (BSC)** and its key performance indicators.

Strategy formulation (Russell & Taylor)

1. **Primary task** — what business are we in?
2. **Core competence** — what do we do better than anyone? (RBV, VRIO)
3. **Winners & qualifiers** — what gets us considered, what wins?
4. **Position** — compete on cost, speed, quality or flexibility

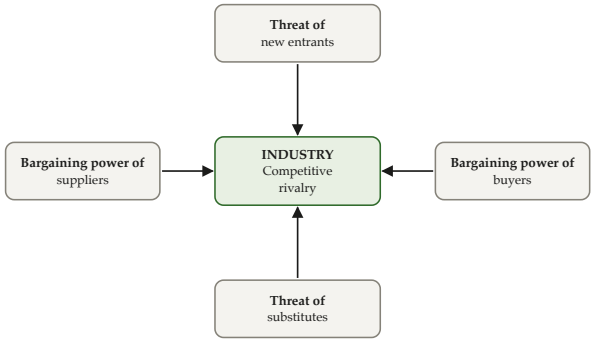


FIGURE 9.1 External analysis: the stronger the four outer forces, the lower the industry's profit potential.

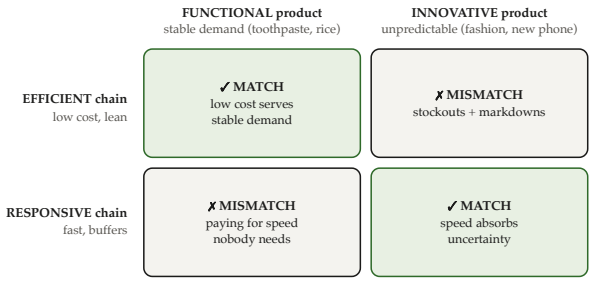


FIGURE 9.2 Fisher's rule: first classify the product by demand uncertainty, then pick the chain. This is the logic behind lean vs agile vs leagile in S7–8.

Inside view and Porter

- **RBV** (Birger Wernerfelt, Jay Barney); **core competence** (Gary Hamel & C.K. Prahalad); VRIO example: Starbucks.
- **Focal firm**: the dominant player that shapes the chain's design and controls resource flows.
- Porter also gave: **value chain analysis**, **generic strategies** (cost leadership, differentiation, focus), **activity mapping** (Southwest Airlines), **Diamond Model** of national competitive advantage.

Competitive priorities with class examples

- **Cost**: waste elimination, lean; low-cost carriers.
- **Speed**: Zara's 9-day design-to-rack.
- **Quality**: Ritz-Carlton's empowered staff.
- **Flexibility**: mass customisation; National Bicycle offers 11,231,862 variations within two weeks.
- Today's winner often becomes tomorrow's qualifier.

Measuring

- **Productivity** = output ÷ input. Single-factor: 1,000 units ÷ 40 labour hours = 25 units/hour. Multifactor = output ÷ (labour + materials + overhead), all in money.
- **Competitiveness**: how well a nation's goods and services pass the test of international markets.
- **BSC** (Robert Kaplan & David Norton): financial, customer, internal process, learning & growth.

Service and industry cases

- **Airlines**: hub-and-spoke vs point-to-point; full-service carrier (FSC) vs low-cost carrier (LCC: lower fares, paid extras, secondary airports); **yield/revenue management** (different prices for the same seat); **overbooking** (sell 105 tickets for 100 seats because some passengers don't show).
- **Service design**: complexity (number of steps) vs divergence (how much each step varies per customer).
- **Focused** operations (Shouldice hernia hospital, KFC) vs **flexible** ones (general hospital).
- **Little's Law**: $L = \lambda W$ (average number in system = arrival rate × time in system).

Other names to recognise: Walmart **cross-docking** (goods move straight from inbound to outbound trucks without storage); **VMI** (Hytrol case); Eliyahu Goldratt's **theory of constraints (TOC)**; Clayton Christensen's **disruptive innovation**; **quality function deployment (QFD)**, which turns customer needs into design specs; **program evaluation and review technique (PERT)**, developed for the US Navy Polaris project; the Halal supply chain (Marko Tieman); Paragon/Wardah's founder values. **History**: craft → division of labour (Adam Smith 1776) → interchangeable parts (Eli Whitney) → scientific management (Frederick Taylor 1911) → moving assembly line (Ford 1913) → quality revolution → lean → ERP and e-commerce → globalisation → sustainability → digital. Keith Oliver of Booz Allen Hamilton coined "supply chain management". **Why go global**: favourable cost, international markets, responding to demand, reliable supply, latest technology.

CHECK YOURSELF

Answer from memory, then check below.

1. **Innovative products with unpredictable demand should use (Fisher)…**
   - A. An efficient supply chain
   - B. A responsive supply chain
   - C. A push system
   - D. A virtual organisation
2. **A bank has 30 customers arriving per hour and each stays 0.2 hours. Average customers inside (Little's Law)?**
   - A. 6
   - B. 15
   - C. 30
   - D. 150
3. RECALL **4 competitive priorities and class examples**

Answers 1. B **2.** A **3.** Cost (lean), speed (Zara 9-day design-to-rack), quality (Ritz-Carlton), flexibility (National Bicycle mass customisation).

CHAPTER 10 · LECTURE NOTES · IMPROVEMENT & QUALITY

Improving operations

PROBLEM
Processes drift, defects appear, customers complain.

→

BIG IDEA
Find the vital few causes, then improve either radically or continuously.

→

HOW (TOOLS)
Four improvement approaches, Pareto, 5 Whys, fishbone, PDCA vs DMAIC.

→

RESULT
Fewer defects, lower cost, happier customers.

IN PLAIN WORDS

There are four broad approaches. **Business process reengineering (BPR)** is radical: throw out the old functional process and rebuild it around the customer (Procter & Gamble's supply chain, Taco Bell's kitchens). **TQM** puts quality at the heart of everything. **Lean**, from the Toyota Production System (TPS), creates smooth flow synchronised with demand. **Six Sigma** is a disciplined, data-driven method to cut defects. They overlap and are often combined.

Improvement can be a big leap (**innovation**/breakthrough) or many small steps (**kaizen**). To know *where* to improve, use a **Pareto chart**: usually about 20% of causes create 80% of problems. Then dig to the root cause with the **5 Whys** or a **fishbone (Ishikawa)** diagram, which groups causes into categories such as people, method, machine and material.

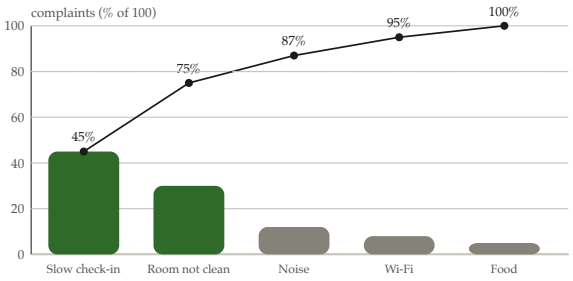


FIGURE 10.1 Sort causes from biggest to smallest and add a cumulative line. The first two bars (green) cause 75% of complaints, so fix those first. Example data, in the spirit of the Hard Rock Hotel Bali case.

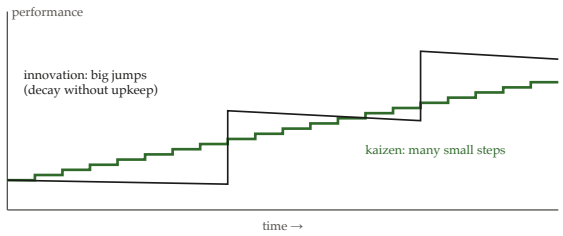


FIGURE 10.2 Kaizen = everyone, small steps, low investment. Innovation/BPR = few champions, big steps, big investment. Best practice combines both.

Innovation vs kaizen

	INNOVATION	KAIZEN
Steps	large, abrupt	small, continuous
Who	few champions	everyone
Investment	large	low
Focus	technology	people and process
Mode	scrap and rebuild	protect and improve

Garvin's 8 dimensions of quality

Performance, features, reliability, conformance, durability, serviceability, aesthetics, perceived quality.

TPS is "lean" because it uses less of everything: inventory, human effort, space, tooling and engineering hours. *Jidoka* (automation with a human touch: stop the line when a defect appears) began on Toyoda looms.

CHECK YOURSELF

Answer from memory, then check below.

1. RECALL Kaizen vs innovation

Answers 1. Kaizen: small, continuous steps, everyone, low investment, people focus. Innovation: large, abrupt steps, few champions, large investment, technology focus.

CHAPTER 11 · FORUM 11 · EXCEL FORECASTING

Forecasting: pick the method, then judge it

PROBLEM
Every plan (production, stock, staff) starts from a demand estimate.

→

BIG IDEA
Let the pattern in the data choose the method; let the error measures judge it.

→

HOW (TOOLS)
Moving averages, simple and multiple regression, seasonal indices, error measures.

→

RESULT
A forecast you can defend with numbers.

IN PLAIN WORDS

Forecasting is the predictive step in analytics. The method depends on the pattern in your past data. With no clear trend, average the recent past using a **moving average (MA)**, or a **weighted moving average (WMA)** that gives newer months more weight. With a steady trend, fit a straight line through time using **simple linear regression**. If demand repeats every year, use **seasonal indices**. If demand depends on several drivers (price, advertising), use **multiple regression**, which Excel runs through its Analysis ToolPak add-in.

Every forecast is wrong by some amount, called the **error** (actual − forecast). You judge methods with three measures. **Mean absolute deviation (MAD)** is the average error size. **Mean squared error (MSE)** squares errors, so big misses count more. **Mean absolute percentage error (MAPE)** gives the error as a percentage, which is easy to compare across products. The method with the lowest values wins.

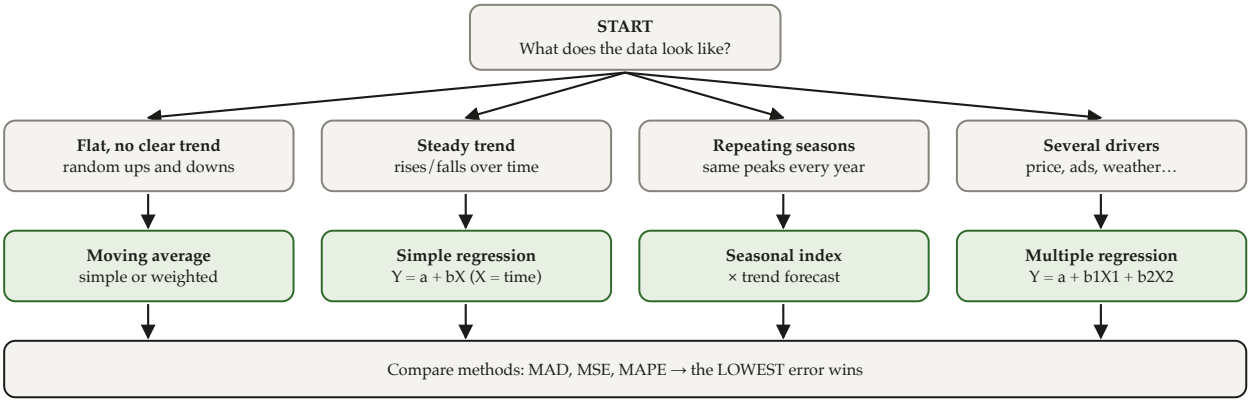


FIGURE 11.1 This is the logical spine of Forum 11: the pattern in the data picks the method, and the error measures judge it.

1 · Simple linear regression

$$\hat{Y} = a + bX$$
$$b = (\Sigma xy - n\cdot\bar{x}\cdot\bar{y}) \div (\Sigma x^2 - n\cdot\bar{x}^2)$$
$$a = \bar{y} - b\cdot\bar{x}$$

Ŷ ("Y-hat") is the forecast, a is the intercept (value when X = 0), b is the slope (change per period), x̄ and ȳ are the averages.

Worked: X = 1–5, Y = 3, 5, 7, 8, 12 → x̄ = 3, ȳ = 7, Σxy = 126, Σx² = 55 → b = 21 ÷ 10 = 2.1, a = 0.7, forecast for X = 6 is 13.3.

Video 1: a = 19.99, b = 1.77 → year 9 ≈ 35.92. Excel: `=SLOPE` `=INTERCEPT` `=FORECAST.LINEAR` .

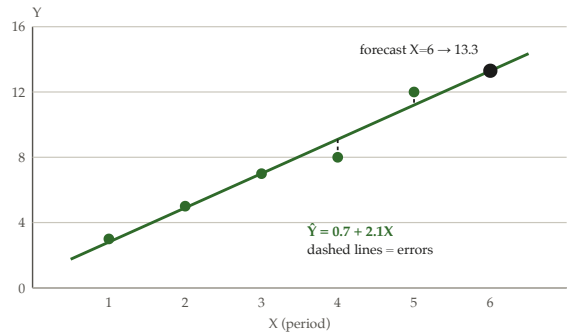


FIGURE 11.2 Least squares picks the one line that makes the sum of the squared dashed gaps as small as possible. b (2.1) = how much Y rises per period; a (0.7) = where the line starts.

2 · Multiple regression: reading the ToolPak output

1 Significance F
< 0.05 → the model as a whole is significant

2 P-value of each X
< 0.05 → keep it; otherwise drop it and rerun

3 R² and adjusted R²
R² (coefficient of determination) = share of Y explained; adjusted R² corrects for the number of Xs

4 Coefficient b
change in Y per +1 unit of that X, others held constant

5 Forecast
plug planned X values into $\hat{Y} = a + b_1X_1 + b_2X_2$

3 · Seasonal index

Index = season average ÷ overall average
Forecast = (annual total ÷ no. of seasons) × index

Q	Y1	Y2	AVG	INDEX	NEXT YR (520)
Q1	80	100	90	0.750	97.5
Q2	120	140	130	1.083	140.8
Q3	150	170	160	1.333	173.3
Q4	90	110	100	0.833	108.3

With a trend: deseasonalise (actual ÷ index) → regression on the result → forecast the trend → reseasonalise (× index). Indices always sum to the number of seasons (4 for quarters).

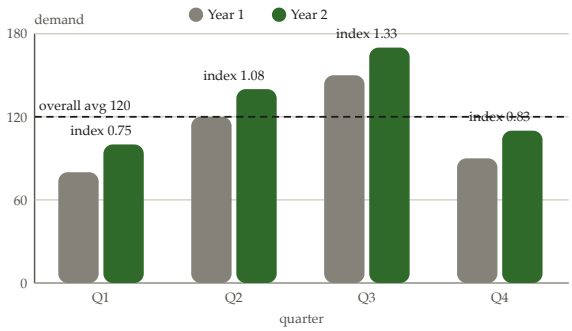


FIGURE 11.3 A seasonal index is just "how far above or below the average line this quarter usually sits". Q3 is 33% above (1.33), Q1 is 25% below (0.75).

4 · Moving averages and error

MA(n) = average of last n actuals
WMA = Σ(weight × actual), weights sum to 1
e = Actual − Forecast
MAD = Σ|e| ÷ n
MSE = Σe² ÷ n
MAPE = Σ(|e| ÷ Actual) ÷ n × 100%

MONTH	ACTUAL	MA3	|E|	E²	|E|/A
4	115	105.00	10.00	100.00	8.70%
5	120	110.00	10.00	100.00	8.33%
6	118	113.33	4.67	21.78	3.95%
Average			MAD 8.22	MSE 73.93	MAPE 6.99%

Month 7: MA3 = (115 + 120 + 118) ÷ 3 = 117.67; WMA with weights 0.5 / 0.3 / 0.2 (newest first) = 118.0.

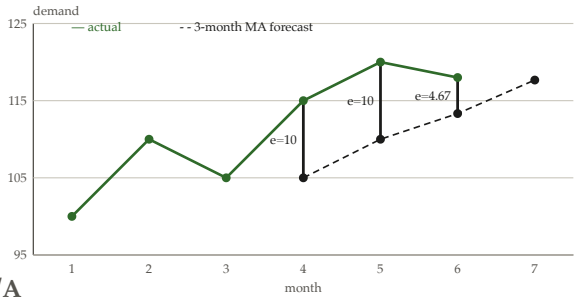


FIGURE 11.4 The dashed MA line always trails the real line when demand is rising, so every error is positive. MAD averages the error sizes, MSE averages their squares, MAPE turns them into percentages.

Exam tip: Lowest MAD/MSE/MAPE wins. A longer MA is smoother but slower to react; a WMA reacts faster to change.

CHECK YOURSELF

Answer from memory, then check below.

1. **Q3 average demand is 160 and the overall quarterly average is 120. The Q3 seasonal index is…**
   - A. 0.75
   - B. 1.20
   - C. 1.33
   - D. 40
2. **Forecast errors of +10, −10 and +4 give a mean absolute deviation (MAD) of…**
   - A. 1.33
   - B. 8.00
   - C. 72.00
   - D. 24.00
3. RECALL **MAD, MSE, MAPE**

Answers 1. C **2.** B **3.** Mean absolute deviation = average |error|; mean squared error = average error²; mean absolute percentage error = average |error| ÷ actual × 100%. Lower is better.

APPENDIX · REFERENCE

Glossary of abbreviations

SHORT	STANDS FOR / MEANING
3PL	Third-Party Logistics provider: a company you hire to run transport/warehousing
4PL	Fourth-Party Logistics provider: integrates and manages the whole chain, often several 3PLs
ACAS	Air Cargo Advance Screening (US)
AEO	Authorised Economic Operator (EU trusted-trader status)
AI / ML	Artificial Intelligence / Machine Learning
BPM	Business Process Management
BPR	Business Process Reengineering
BSC	Balanced Scorecard
CILO	Course Intended Learning Outcome
CO_2eq	Carbon dioxide equivalent: other gases converted to the same warming effect as CO_2
CSI	Container Security Initiative (US)
CSR	Corporate Social Responsibility
C-TPAT	Customs-Trade Partnership Against Terrorism (US)
DC	Distribution Centre
DMAIC	Define, Measure, Analyse, Improve, Control (Six Sigma cycle)
ERP	Enterprise Resource Planning: one integrated company-wide software system
ESG	Environmental, Social and Governance
FP	Foundational Premise
FSC / LCC	Full-Service Carrier / Low-Cost Carrier (airlines)
FTZ	Free Trade Zone
FX	Foreign Exchange
GHG	Greenhouse Gas
HRM	Human Resource Management
ICD	Inland Container Depot: inland "dry port" for customs clearance

IMO	International Maritime Organization
Incoterms	International Commercial Terms: who pays and carries risk at each point of delivery
IoT	Internet of Things: connected sensors
ISO	International Organization for Standardization
ISPS	International Ship and Port Facility Security code
JIT	Just-In-Time
KPI	Key Performance Indicator
L/C	Letter of Credit
LSCM	Logistics and Supply Chain Management
MA / WMA	Moving Average / Weighted Moving Average
MAD	Mean Absolute Deviation
MAPE	Mean Absolute Percentage Error
MaaS	Mobility as a Service
MIS	Management Information Systems
MSE	Mean Squared Error
MTO / MTS	Make-To-Order / Make-To-Stock
OM	Operations Management
PDCA	Plan, Do, Check, Act (Deming cycle)
PERT	Program Evaluation and Review Technique
POS	Point Of Sale: checkout sales data
QFD	Quality Function Deployment
R^2	Coefficient of determination: share of variation in Y explained
RBV	Resource-Based View
SCaaS	Supply Chain as a Service
SCM	Supply Chain Management
SCRM	Supply Chain Risk Management
SKU	Stock-Keeping Unit: one specific product variant
SLA	Service Level Agreement

TOC	Theory of Constraints
TPS	Toyota Production System
TQM	Total Quality Management
UCP 600	Uniform Customs and Practice for Documentary Credits, publication 600: the international rules for letters of credit
VMI	Vendor-Managed Inventory: the supplier monitors and refills the customer's stock
VRIO	Valuable, Rare, Inimitable, Organised
VUCA	Volatility, Uncertainty, Complexity, Ambiguity
XaaS	Anything as a Service

MGMT6564 · UTS (MIDTERM) · KISI-KISI

UTS Kisi-Kisi: 14 questions

Each of the lecturer's 14 kisi-kisi questions, reasoned step by step.
Sources: *course slides (Sessions 1–14), Russell & Taylor Operations and Supply Chain Management ch. 1, Mr. Syamil's lecture notes, Forum 11 (forecasting), and the lecturer's kisi-kisi.*

How to use this report to study

Every answer in this report follows the same chain of thought. If you learn the chain, you can answer questions you have never seen:

1. **Why does this exist?** Name the business problem first.
2. **What is it?** Give the definition in one sentence.
3. **How does it work?** Break it into parts or steps.
4. **Show it.** Draw the diagram.
5. **Prove it.** Give a real example.
6. **So what?** Evaluate: when it helps, when it doesn't, which option is better.

Lecturers mark the *reasoning*, not just the keywords. A short answer that walks through these six moves usually beats a long list of facts.

QUESTION 1 · KISI-KISI

Linear Programming (LP): define it and give real business cases

WHAT THE QUESTION REALLY ASKS

Explain what Linear Programming is, why managers use it, how a problem is built, and give real business cases. A small worked example shows you truly understand it.

Keywords to hit: optimisation, decision variables, objective function, constraints, limited resources, feasible region, corner point, Excel Solver.

THINKING IT THROUGH, STEP BY STEP

1 What problem do managers face?

Every firm has **limited resources**: machine hours, labour, money, truck space, raw material. At the same time it has **many possible choices**: which products to make, how many, from which factory to ship. Guessing wastes money. The manager needs the single best combination.

2 So what is LP?

Linear Programming (LP) is a mathematical method that finds the **best (optimal) decision**, either the maximum profit or the minimum cost, **while respecting all the limits (constraints)**. It is called *linear* because the objective and every constraint are straight-line equations (no squares, no multiplying two variables together). It sits in **prescriptive analytics**: it does not just describe or predict, it tells you *what to do*.

3 What are the building blocks?

Every LP model has four parts:

1. **Decision variables**: what you control (e.g. T = tables per week, C = chairs per week).
2. **Objective function**: what you maximise or minimise (e.g. maximise profit $Z = 70T + 50C$).
3. **Constraints**: the limits (e.g. carpentry hours ≤ 240).
4. **Non-negativity**: you cannot make a negative number of products ($T, C \geq 0$).

4 **What must be true for LP to work? (assumptions)**

- **Certainty**: the numbers (profit per unit, hours per unit) are known and fixed.
- **Proportionality**: double the output, double the resource used and the profit earned.
- **Additivity**: the total is the sum of the parts (no interaction effects).
- **Divisibility**: answers may be fractions (if you need whole numbers you use integer programming).

5 **How is it solved?**

With two variables you can graph it. Each constraint is a line; the area that satisfies all of them is the **feasible region**. The optimum always lies on a **corner point** of that region, so you only need to test the corners. With many variables, managers use **Excel Solver** (set objective cell, variable cells, constraints, choose "Simplex LP").

6 **Worked example (classic furniture problem).**

A workshop makes tables (profit $70) and chairs (profit $50).

- Carpentry: table 4 h, chair 3 h, **240 h available** → $4T + 3C \leq 240$
- Painting: table 2 h, chair 1 h, **100 h available** → $2T + C \leq 100$

Test the corner points:

CORNER (T, C)	PROFIT 70T + 50C
(0, 0)	$0
(0, 80)	$4,000
(30, 40)	**$4,100 ← optimum**
(50, 0)	$3,500

How (30, 40) was found: solve both constraints together. From painting, $C = 100 - 2T$. Put it into carpentry: $4T + 3(100 - 2T) = 240 \rightarrow 4T + 300 - 6T = 240 \rightarrow T = 30$, so $C = 40$.

Decision: make 30 tables and 40 chairs per week for a maximum profit of $4,100. Both resources are fully used (binding constraints).

7 Where is LP used in business and supply chains?

BUSINESS PROBLEM	VARIABLES	OBJECTIVE	CONSTRAINTS
Product mix (factory deciding what to make)	units of each product	max profit	machine hours, labour, materials
Transportation problem (ship from factories to warehouses)	units shipped on each route	min shipping cost	factory capacity, warehouse demand
Blending / diet (animal feed, fuel blending, food recipes)	kg of each ingredient	min ingredient cost	nutrition or quality specs
Workforce scheduling (call centre, hospital, airline crew)	staff per shift	min labour cost	staff needed each hour
Network / facility design	which warehouse serves which region	min total logistics cost	capacity, service level
Media / marketing budget	ads in each channel	max reach	budget, minimum exposure

ILLUSTRATION

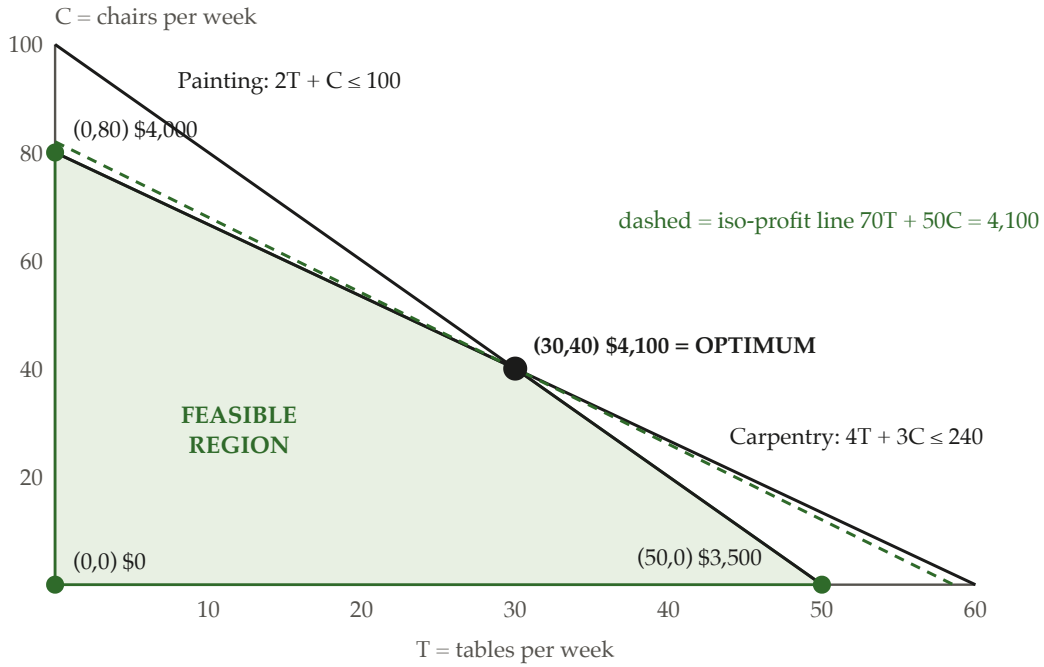


FIGURE 1.1 *Each constraint is a line. The green area satisfies all of them. The dashed iso-profit line slides outwards until it last touches the region, at the corner (30, 40).*

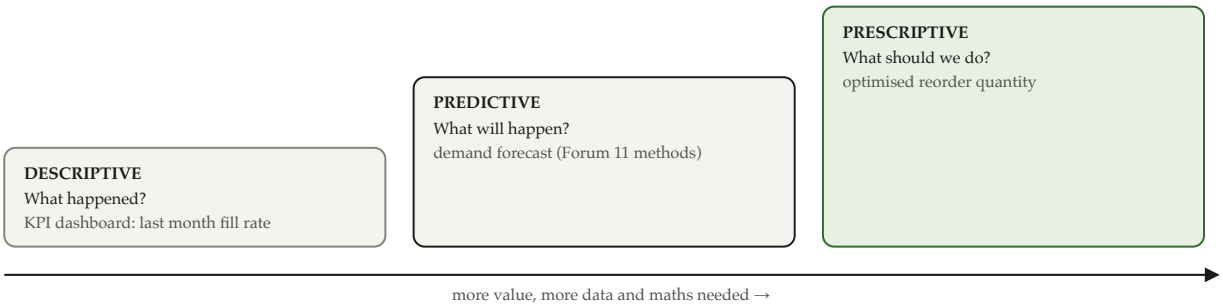


FIGURE 1.2 *LP is prescriptive analytics: it answers "what should we do?".*

Draw this in the exam: two axes (T and C), two constraint lines, shade the feasible region, mark the four corners with their profit, and circle the optimum.

EXAM-READY ANSWER

MODEL ANSWER

Linear Programming (LP) is a mathematical optimisation technique that helps managers allocate **limited resources** to reach the **best outcome**, maximum profit or minimum cost, when the objective and all constraints are linear. An LP model has four parts: **decision variables** (what we choose), an **objective function** (what we maximise or minimise), **constraints** (limits such as hours, budget or capacity) and **non-negativity**. It assumes certainty, proportionality, additivity and divisibility. The set of solutions meeting all constraints is the **feasible region**, and the optimum always lies on one of its corner points. Small problems can be solved graphically; real ones use Excel Solver.

Example: a workshop earns \$70 per table and \$50 per chair, with 240 carpentry hours (4 h and 3 h each) and 100 painting hours (2 h and 1 h each). Testing the corners (0,80) = \$4,000, (30,40) = \$4,100 and (50,0) = \$3,500 shows the best plan is 30 tables and 40 chairs, for \$4,100.

Business cases: **product mix** in a factory; the **transportation problem** (cheapest way to ship from plants to warehouses); **blending** (animal feed or fuel at minimum cost while meeting quality specs); **staff scheduling** in hospitals, call centres and airlines; and **distribution network design**. In supply chains, LP turns data into a concrete decision, which is why it is called prescriptive analytics.

COMMON MISTAKES

- ✗ Defining LP as "a forecasting method". It is an **optimisation** method; forecasting only provides its input numbers.
- ✗ Forgetting non-negativity or the units of each constraint.
- ✗ Choosing the corner by eye. Always compute the profit at every corner.
- ✗ Listing examples without saying what is maximised or minimised and what the constraints are.

QUESTION 2 · KISI-KISI

Business and supply chain problems from the BINUS Forum

WHAT THE QUESTION REALLY ASKS

Describe real problems discussed in the course forum and show how a supply chain tool solves them. The lecturer wants to see that you can move from a messy business problem to a method and then to a decision.

Keywords to hit: the problem, the data, the tool, the result, the managerial decision.

> **Honesty note:** this report can only use the forum material that is in the project notes, which is **Forum 11 (forecasting in Excel)**. If you answered other forum topics (for example a Solver/LP case), use the same 4-step structure below with your own forum post.

THINKING IT THROUGH, STEP BY STEP

1. **Why do forum problems matter?**

   Every supply chain plan (how much to produce, how much stock to hold, how many staff to roster) starts from **an estimate of the future**. If the estimate is wrong, the whole chain is wrong: too much stock (cost) or too little (lost sales). That is exactly what Forum 11 practised.

2. **The answering pattern for any forum problem.**

   1. **Problem**: what decision is the business stuck on?
   2. **Data pattern**: what does the history look like?
   3. **Tool**: which method fits that pattern?
   4. **Decision**: what number do we get, and what should the manager do with it?

3. **Choose the tool from the data pattern (this is the key logic).**

   - Flat, random ups and downs → **moving average**
   - A steady trend over time → **simple linear regression** ($Y = a + bX$)
   - The same peaks every season → **seasonal index** × trend
   - Several drivers (price, advertising, weather) → **multiple regression**
   - Then compare methods with error measures: **MAD** (Mean Absolute Deviation), **MSE** (Mean Squared Error), **MAPE** (Mean Absolute Percentage

Error). **The lowest error wins.**

4 **The four Forum 11 problems, each in the pattern.**

#	BUSINESS PROBLEM	TOOL	RESULT / DECISION
1	**Rising cost per unit**: management wants to budget next year's unit cost	Simple linear regression on time (Excel `=SLOPE`, `=INTERCEPT`, `=FORECAST.LINEAR`)	a = 19.99, b = 1.77 → year 9 cost ≈ 19.99 + 1.77 × 9 ≈ **35.92**. Budget for cost to rise about 1.77 per year.
2	**What drives sales?** (e.g. price and advertising)	Multiple regression (Excel Data Analysis ToolPak)	Check **Significance F < 0.05** (model works), **each P-value < 0.05** (keep that driver), **R^2** (share of sales explained). Spend on the drivers that matter.
3	**Seasonal demand**: peaks in the same quarters each year	Seasonal index = season average ÷ overall average, then trend forecast × index	Plan capacity, staff and stock ahead of the peak instead of reacting to it.
4	**Noisy demand with no trend**	3-period moving average, judged by MAD/MSE/MAPE	Smooths random noise; but it **lags** a trend, so it is not used when demand is clearly rising.

5 **Connect it back to supply chain management.**

Better forecasts → fewer surprises → less safety stock and less **bullwhip effect** (Q5). That is why forecasting is a supply chain topic, not just a statistics topic.

ILLUSTRATION

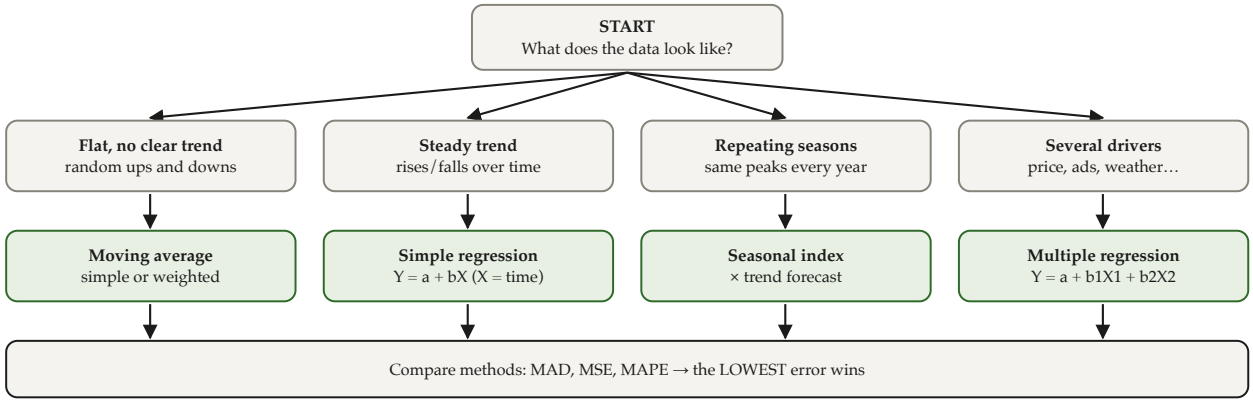


FIGURE 2.1 *Let the shape of the data choose the method; let the error measures judge it.*

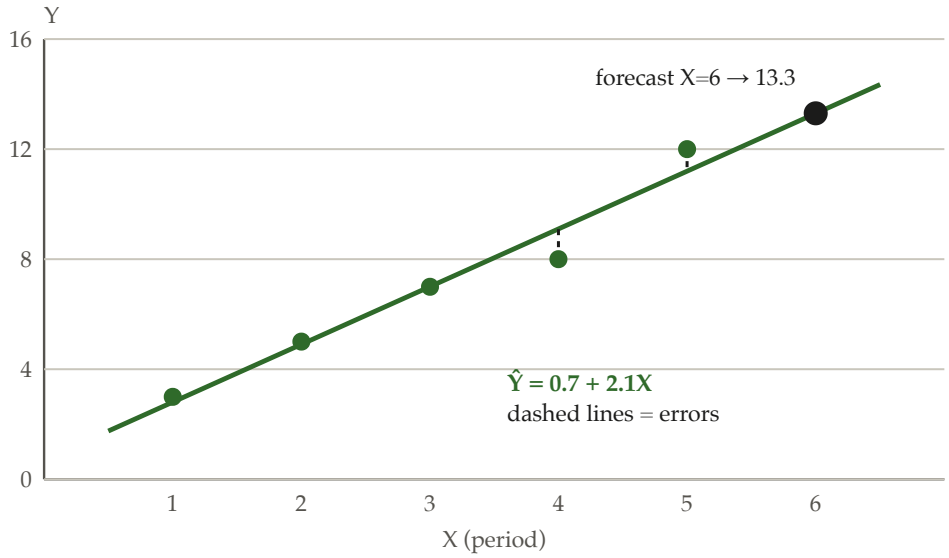


FIGURE 2.2 *Worked mini-example: X = 1–5, Y = 3, 5, 7, 8, 12 → $\hat{Y}$ = 0.7 + 2.1X; forecast for X = 6 is 13.3.*

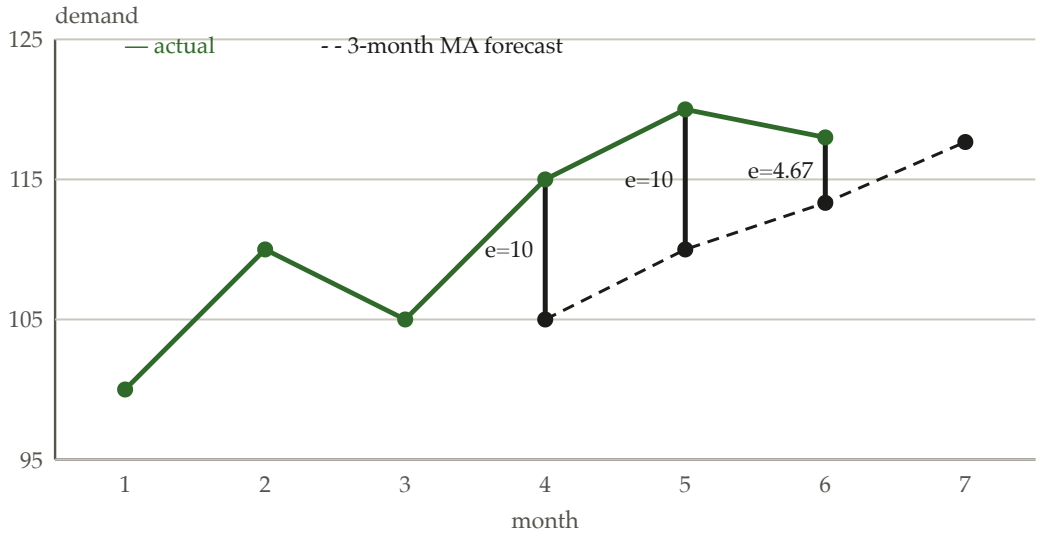


FIGURE 2.3 *The dashed moving average always trails the actual line. The vertical gaps are the errors that MAD, MSE and MAPE average.*

Draw this in the exam: a small scatter of points with a straight trend line, and label "a = intercept, b = slope, forecast = a + bX".

EXAM-READY ANSWER

MODEL ANSWER

In BINUS Forum 11 we worked on forecasting problems, because every supply chain decision (production, inventory, staffing) depends on an estimate of future demand or cost.

(1) Rising unit cost. A firm wanted to budget its future cost per unit. Because the data showed a steady upward trend, we used **simple linear regression** ($Y = a + bX$). Excel gave $a = 19.99$ and $b = 1.77$, so the forecast for year 9 is about **35.92**: the cost rises about 1.77 per year and the budget should reflect that.

(2) Sales drivers. To learn what drives sales, we used **multiple regression** in the Data Analysis ToolPak. The model is valid if Significance F < 0.05, each driver is kept only if its P-value < 0.05, and R^2 tells how much of sales is explained.

(3) Seasonal demand. Where demand peaks in the same seasons, a **seasonal index** (season average ÷ overall average) is multiplied by the trend forecast, so capacity and stock are prepared before the peak.

(4) Noisy demand. Where demand has no trend, a **moving average** smooths the noise. Methods are compared with **MAD, MSE and MAPE**, and the method with the lowest error is chosen.

The lesson: diagnose the data pattern first, choose the matching tool, and measure the error before trusting the forecast. Accurate forecasts reduce safety stock and the bullwhip effect.

COMMON MISTAKES

- ✗ Naming a method without saying *why* it fits the data pattern.
- ✗ Using a moving average on trending data (it lags behind).
- ✗ Reading R^2 as "the model is significant". Significance comes from Significance F and the P-values.
- ✗ Stopping at the number. Always end with the managerial decision.

QUESTION 3 · KISI-KISI

Define Operations Management (OM)

WHAT THE QUESTION REALLY ASKS

Give a precise definition of Operations Management, explain the idea behind it (transformation and value added), and show its place in the firm.

Keywords to hit: design, operation, improvement, productive system, inputs → transformation → outputs, value added, efficiency and effectiveness.

THINKING IT THROUGH, STEP BY STEP

1. **Why does a firm need OM?**

   Every organisation exists to **deliver something customers value**: a product, a service or both. Somebody must decide *how* that thing is made and delivered: the process, the capacity, the quality, the schedule, the stock. That "how" is operations.

2. **The definition.**

   Russell & Taylor: **Operations Management (OM) is the design, operation and improvement of productive systems.** A productive system is any system that turns inputs into goods or services.

   - *Design*: choose the product, process, capacity and location (long-term decisions).
   - *Operation*: plan, schedule, control quality and inventory day to day.
   - *Improvement*: keep making it better (lean, Six Sigma, kaizen).

3. **The core idea: transformation creates value.**

   OM transforms **inputs** (materials, machines, labour, management, capital) into **outputs** (goods and services) **of greater value**.

   Value added = value of outputs − cost of inputs. If the output is not worth more than the inputs, the operation destroys value.

4. **How do we know OM is doing well?**

   - **Efficiency**: doing things right, measured as **productivity = output ÷ input**.

- **Effectiveness**: doing the right things, i.e. what customers actually want (cost, quality, speed, flexibility).

5 **Where does OM sit in the company?**

OM is one of the three core functions, beside **marketing** (creates demand) and **finance** (provides money). OM *fulfils* the demand. In a modern firm, OM extends beyond the factory walls into the **supply chain** (suppliers and distributors), which is why this course joins OM and supply chain management.

ILLUSTRATION

OM is illustrated in Q4 (`figures/q04-om-transformation.svg`). For Q3, a one-line sketch is enough: **Inputs → Transformation → Outputs (greater value)**, with a feedback arrow underneath.

EXAM-READY ANSWER

MODEL ANSWER

Operations Management (OM) is the **design, operation and improvement of the productive systems** that create a firm's goods and services. Its core is the **transformation process**: inputs such as materials, machines, labour, management and capital are transformed into outputs (goods and services) of **greater value** than the inputs. The difference is the **value added**. OM makes decisions on design (product, process, capacity, location), on daily operation (scheduling, quality, inventory) and on continuous improvement. It is judged by **efficiency** (productivity = output ÷ input) and **effectiveness** (meeting customer needs on cost, quality, speed and flexibility). OM is one of the three core business functions, together with marketing and finance, and today it extends across the whole supply chain.

COMMON MISTAKES

✗ "OM is about factories." Services (hospitals, banks, universities, airlines) are productive systems too.

✗ Leaving out "value added", which is the reason OM exists.

✗ Confusing efficiency with effectiveness.

QUESTION 4 · KISI-KISI

Illustrate Operations Management (OM)

WHAT THE QUESTION REALLY ASKS

Draw the transformation model and explain it with a concrete example. They want a picture plus words.

Keywords to hit: inputs, transformation process, outputs, feedback/control, types of transformation, value added.

THINKING IT THROUGH, STEP BY STEP

1. **Start with the three boxes.**

   Inputs → Transformation process → Outputs. This shows the *flow*.

2. **Fill the boxes.**

   - Inputs: materials, machines and equipment, labour, management, capital.
   - Transformation: what the operation does to the inputs.
   - Outputs: goods and services worth more than the inputs.

3. **Add the feedback loop.**

   Managers **measure** the outputs (quality, cost, delivery time) and **adjust** the inputs and the process. Without this loop there is no control and no improvement.

4. **Show that "transformation" has six forms**

   (so the diagram works for services too):

TYPE	WHAT CHANGES	EXAMPLE
Physical	the shape or form	factory making noodles
Locational	the place	truck, airline, courier
Exchange	the ownership	retail store, e-commerce
Physiological	the body	hospital, gym
Psychological	the feelings	cinema, concert
Informational	the information	telecom, news, bank data

5 Walk one example through the diagram.

A bakery: inputs are flour, sugar, ovens, bakers and cash. The transformation is mixing and baking (physical). The output is bread that customers pay more for than the ingredients cost. Feedback: complaints about burnt bread lead to adjusted oven settings.

BINUS University: inputs are students, lecturers, buildings and LMS. The transformation is teaching (informational and psychological). The output is graduates with skills. Feedback comes from grades and graduate employment.

ILLUSTRATION

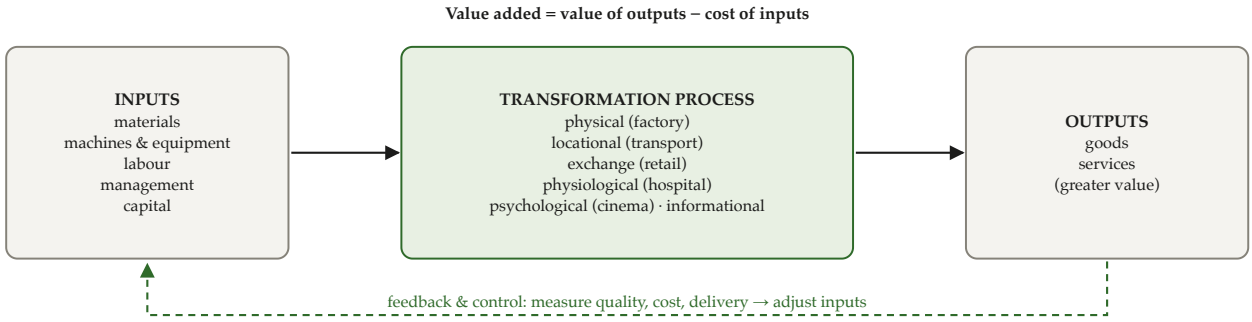


FIGURE 4.1 *Value added = value of outputs − cost of inputs. The dashed feedback loop is how managers control and improve the process.*

Draw this in the exam: three boxes left to right with arrows, list inputs and outputs inside, write the six transformation types in the middle box, and draw a dashed feedback arrow from outputs back to inputs labelled "measure & adjust".

EXAM-READY ANSWER

MODEL ANSWER

OM can be illustrated as a **transformation model**: **inputs** (materials, machines, labour, management, capital) enter a **transformation process** and leave as **outputs** (goods and services) of greater value. A **feedback and control loop** measures output performance (quality, cost, delivery) and adjusts the inputs and the process. The transformation may be **physical** (manufacturing), **locational** (transport), **exchange** (retail), **physiological** (healthcare), **psychological** (entertainment) or **informational** (telecom). *Example:* a bakery turns flour, ovens and bakers' labour into bread through physical

transformation; customer complaints feed back to adjust baking. The **value added** is the bread's value minus the cost of its inputs.

(Draw the three boxes, arrows and the dashed feedback loop.)

COMMON MISTAKES

✗ Drawing only three boxes with no feedback loop.

✗ Giving only a factory example. Add one service example to show you understand the six types.

QUESTION 5 · KISI-KISI

The bullwhip effect: explain it, with an example and an illustration (Hau Lee)

WHAT THE QUESTION REALLY ASKS

Explain what the bullwhip effect is, why it happens (Hau Lee's four causes), why it is harmful, how to reduce it, and draw it. This was **specially flagged by the lecturer**, so expect high marks on it.

Keywords to hit: order variability amplifies upstream, demand forecast updating, order batching, price fluctuation, rationing and shortage gaming, information sharing, POS, VMI, EDLP.

THINKING IT THROUGH, STEP BY STEP

1. **Notice the strange fact.**

   Consumers buy fairly steadily: people use about the same number of diapers or bottles of water every week. Yet factories see **wild swings** in the orders they receive. Where does the swing come from, if not from the consumer?

2. **Name it.**

   The **bullwhip effect** is the **amplification of order variability as you move upstream** in the supply chain: from the retailer to the wholesaler to the distributor to the manufacturer and its suppliers. Like cracking a whip, a small flick of the wrist (consumer) becomes a big swing at the tip (factory). The term comes from **Hau Lee, Padmanabhan and Whang (1997), "The Bullwhip Effect in Supply Chains."**

3. **Make it concrete with numbers (the lecturer's sketch).**

   A small change in consumer demand of **5** units becomes **10** at the retailer's order, **15** at the wholesaler's order and **20** at the factory. Each player adds a bit of "just in case".

4. **Ask why each player amplifies: Hau Lee's four causes.**

   1. **Demand forecast updating.** Each level forecasts from the *orders* it receives, not from real consumer sales, and adds its own safety stock. Forecasts stacked on forecasts magnify the swing.

2. **Order batching.** Firms order in big weekly or monthly lots (to save ordering or transport cost), so smooth demand looks lumpy upstream.
3. **Price fluctuation.** Promotions and discounts make buyers **forward-buy** (stock up while it is cheap, then stop ordering).
4. **Rationing and shortage gaming.** When supply is short and allocated, buyers **inflate orders** to get a bigger share, then cancel when supply recovers. The factory sees "phantom demand".

5 **Why is this bad?**

The factory plans to the exaggerated orders, so the chain swings between **excess inventory** and **stockouts**. Result: wasted capacity, overtime followed by idle time, higher transport and storage cost, worse customer service and greater risk to the whole system.

6 **So the cure follows from the cause (cause → countermeasure).**

CAUSE	COUNTERMEASURE
Demand forecast updating	Share **POS** (point-of-sale) data with every level; **VMI** (vendor-managed inventory); shorter lead times
Order batching	Smaller, more frequent orders; electronic ordering; mixed truckloads through a **3PL** (third-party logistics provider)
Price fluctuation	**EDLP** (everyday low price); fewer promotions
Rationing and shortage gaming	Allocate by past sales, not by orders; share capacity and stock information; penalise cancellations

The common thread: **share real information and reduce the incentives to over-order.**

7 **Real examples.**

- **Procter & Gamble's Pampers diapers**: babies use diapers at a steady rate, yet P&G saw big swings in distributor orders. This is the classic case behind Lee's article.
- **Toilet paper in early 2020 (COVID-19)**: panic buying plus retailers over-ordering created empty shelves, then excess stock months later.
- **Car chip shortage 2021**: carmakers cancelled chip orders, then panic-ordered; suppliers saw huge swings.

- **The Beer Game** (MIT simulation) shows the effect even when players are smart, because each sees only local orders.

ILLUSTRATION

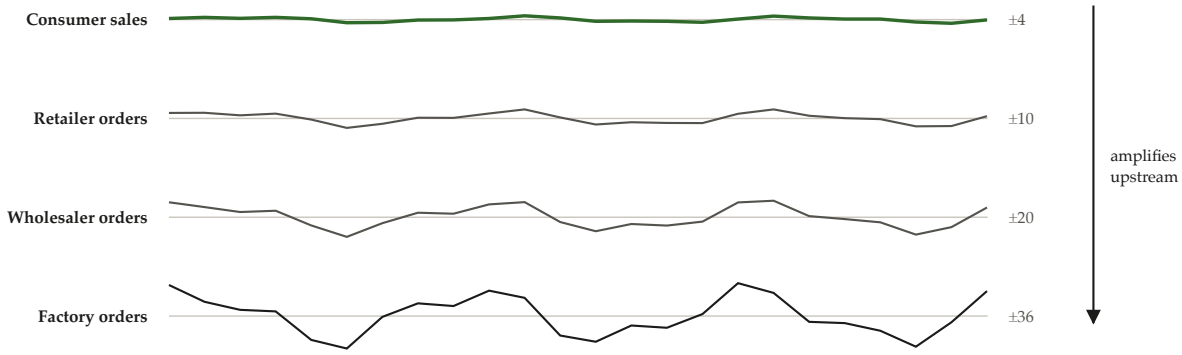


FIGURE 5.1 *The same small wobble in consumer sales (±4) grows into ±36 in factory orders. Each tier adds its own forecasting and safety stock.*

Draw this in the exam (lecturer's version): four small boxes in a row, Consumer → Retailer → Wholesaler → Factory, with a wave above each one that gets taller. Write **5 → 10 → 15 → 20** under them and an arrow labelled "variability amplifies upstream". Then list the four causes beside it.

EXAM-READY ANSWER

MODEL ANSWER

The **bullwhip effect** is the **amplification of order variability as orders move upstream** in the supply chain, from consumer to retailer, wholesaler and manufacturer (Hau Lee, Padmanabhan and Whang, 1997). A small change in consumer demand, say 5 units, becomes an order change of 10 at the retailer, 15 at the wholesaler and 20 at the factory, even though real consumption is stable.

Causes (Hau Lee): (1) **demand forecast updating**: each tier forecasts from the orders it receives and adds safety stock; (2) **order batching**: large periodic orders make demand look lumpy; (3) **price fluctuation**: promotions cause forward buying; (4) **rationing and shortage gaming**: during shortages buyers inflate orders and later cancel.

Why it is bad: excess inventory alternating with stockouts, wasted capacity, overtime then idle time, higher costs, poorer service and higher risk across the chain.

Countermeasures: share POS data and use VMI (vendor-managed inventory); order smaller and more often, using electronic ordering and mixed truckloads via a 3PL; use EDLP (everyday low price) instead of frequent promotions; allocate scarce supply by past sales and penalise cancellations.

Example: Procter & Gamble saw large swings in distributor orders for Pampers diapers although babies' consumption was steady; similar swings hit toilet paper during COVID-19 panic buying.

(Draw the chain with growing waves and 5 → 10 → 15 → 20.)

COMMON MISTAKES

- ✗ Saying the bullwhip effect is "high consumer demand". It is about **variability growing upstream**, not demand being high.
- ✗ Drawing the waves getting *smaller* towards the factory (they grow).
- ✗ Listing causes without the matching countermeasures. Pair them.
- ✗ Forgetting to name Hau Lee, which the lecturer specifically mentioned.

QUESTION 6 · KISI-KISI

Supply Chain Management (SCM): define it, illustrate it, and explain 1PL–5PL and the value chain

WHAT THE QUESTION REALLY ASKS

Give the definition of SCM, draw the chain with its flows, and explain the related ideas: the levels of logistics outsourcing (1PL to 5PL), Porter's value chain, and the foundational premises.

Keywords to hit: network of organisations, upstream/downstream, three flows (materials, information, money), focal firm, 1PL–5PL, primary and support activities, margin.

THINKING IT THROUGH, STEP BY STEP

1. **Why did SCM appear?**

   Products today pass through **many companies** before reaching the consumer (raw material → components → assembly → distribution → retail). A firm can be perfect internally and still fail if a supplier is late or a distributor holds too much stock. So **firms no longer compete alone; supply chains compete against supply chains.**

2. **Definition.**

   Supply Chain Management (SCM) is the management of the flow of **materials, information and money** across the **entire network** of organisations, from raw-material suppliers through manufacturers and distributors to the final consumer, in order to deliver customer value at the lowest total cost.

   (Council of Supply Chain Management Professionals, CSCMP: SCM covers the planning and management of all activities in sourcing and procurement, conversion and logistics, including coordination and collaboration with channel partners.)

3. **Picture the chain and its three flows.**

   - **Materials** flow **downstream** (supplier → consumer); returns flow back (reverse logistics, Q14).
   - **Information** flows **both ways** (orders up; forecasts, POS data and shipping notices both directions).

- **Money** flows **upstream** (consumer pays retailer, retailer pays distributor, and so on).

The firm you look from is the **focal firm**. Everything before it is **upstream** (suppliers: tier 1, tier 2), everything after it is **downstream** (customers).

4 **How does a chain behave? The six foundational premises (FP) from the slides.**

1. FP1 **Network**: a chain is nodes (firms) and links (flows).
2. FP2 **Complex**: a complex adaptive system; there is tension between control and emergence.
3. FP3 **Relative**: a chain is always defined from one product and one agent's point of view.
4. FP4 **Two layers**: the physical chain plus a support chain (banks, IT, logistics providers).
5. FP5 **Limited view**: bounded by what the focal firm can see.
6. FP6 **Fading view**: visibility fades with physical distance, cultural distance and closeness.

In one sentence: *a supply chain is a network no single firm fully controls or sees.*

5 **Who moves the goods? The 1PL → 5PL ladder (logistics outsourcing).**

The logic: each step up, the firm **hands more of the logistics to outsiders** and gains more integration.

LEVEL	MEANING	EXAMPLE
1PL First-Party Logistics	The firm does it itself: own trucks, own warehouse	A farmer delivering their own harvest; a factory with its own fleet
2PL Second-Party Logistics	Hires a **carrier** for one leg of transport	Trucking company, shipping line, airline cargo
3PL Third-Party Logistics	Outsources **logistics operations**: warehousing + transport + value-added services (packing, labelling)	DHL Supply Chain, Kerry Logistics, JNE/SiCepat fulfilment
4PL Fourth-Party Logistics	An **integrator** that manages the whole chain and several 3PLs, as one point of contact	Lead logistics provider / consulting-style integrator

LEVEL	MEANING	EXAMPLE
5PL Fifth-Party Logistics	Manages **whole networks** for many shippers, often through technology platforms (aggregating demand, e-commerce)	E-commerce logistics platforms

6 **Where does value come from? Porter's value chain (1985).**

Michael Porter split a firm into activities that create value:

- **Primary activities** (the flow of the product): **inbound logistics → operations → outbound logistics → marketing & sales → service**.
- **Support activities** (help all primary ones): **firm infrastructure, human resource management, technology development, procurement**.
- **Margin** = the value customers pay for minus the cost of all activities.

Link to SCM: **inbound and outbound logistics are inside every firm's value chain**, and the supply chain is simply many value chains linked together (sometimes called the *value system*).

7 **So what?**

Good SCM raises customer value (right product, on time) and lowers total cost (less stock, less waste), and it reduces risks such as the bullwhip effect.

ILLUSTRATION

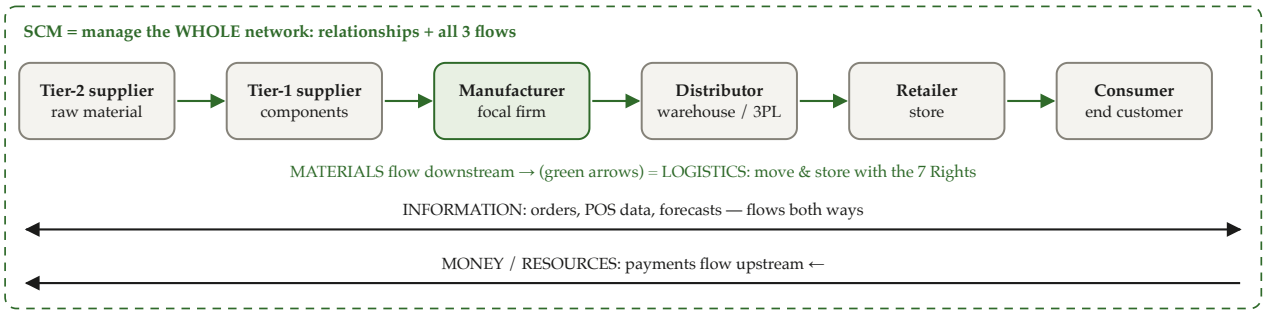


FIGURE 6.1 *SCM manages the whole network and all three flows. The green material links are logistics.*

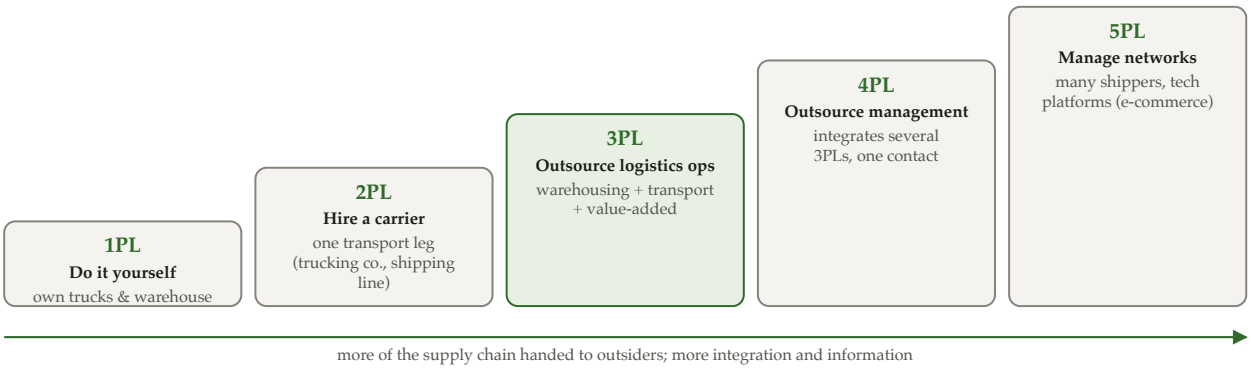


FIGURE 6.2 *Each step up hands more of the chain to outsiders, with more integration and information.*

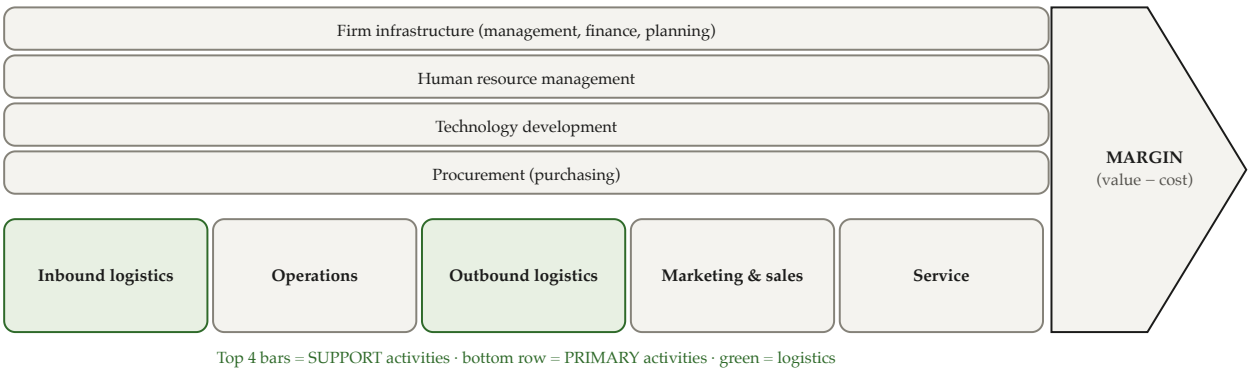


FIGURE 6.3 *Four support activities sit above five primary activities. Green marks the two logistics activities. Margin is what is left after all costs.*

Draw this in the exam: a row of boxes Supplier → Manufacturer → Distributor → Retailer → Consumer; one arrow below for materials (→), a two-headed arrow for information (↔), and one for money (←). Then a staircase 1PL–5PL. If asked about the value chain, draw Porter's arrow shape.

EXAM-READY ANSWER

MODEL ANSWER

Supply Chain Management (SCM) is the coordination of the **flows of materials, information and money** across the **whole network** of organisations, from raw-material suppliers through manufacturers, distributors and retailers to the final consumer, to deliver customer value at the lowest total cost. Materials flow downstream, money flows upstream and information flows both ways. Because no single firm controls or sees the whole network (the six foundational premises: network, complex, relative, two layers, limited view,

fading view), firms must collaborate; today *supply chains compete against supply chains*.

Logistics can be kept in-house or outsourced at increasing levels: **1PL** (own trucks and warehouses), **2PL** (hire a carrier for transport), **3PL** (outsource warehousing, transport and value-added services), **4PL** (an integrator managing several 3PLs and the whole chain) and **5PL** (managing whole networks for many shippers, often via technology platforms).

Porter's **value chain** explains where value is created inside each firm: five **primary activities** (inbound logistics, operations, outbound logistics, marketing and sales, service) supported by four **support activities** (firm infrastructure, human resource management, technology development, procurement), producing a **margin**. A supply chain links the value chains of many firms.

Example: a bottle of Aqua water involves a plastic maker, a bottling plant, a 3PL distributor, retailers and consumers, plus the payments and order information flowing between them.

COMMON MISTAKES

- ✗ Saying SCM = logistics. Logistics is only part of SCM (see Q7).
- ✗ Drawing only the material flow. Always draw all three flows with their directions.
- ✗ Mixing up 2PL (a carrier) and 3PL (outsourced logistics operations).
- ✗ Putting procurement as a primary activity in Porter's chain. It is a **support** activity.

QUESTION 7 · KISI-KISI

The relationship between logistics and Supply Chain Management (SCM)

WHAT THE QUESTION REALLY ASKS

Explain how logistics and SCM are connected and how they differ. The best answers show that logistics is a part of SCM, with a nested diagram.

Keywords to hit: subset, physical flow and storage, 7 Rights, forward and reverse, relationships, all three flows, strategic vs operational.

THINKING IT THROUGH, STEP BY STEP

1. **What does logistics do?**

   Logistics plans, carries out and controls the efficient **forward and reverse flow and storage of goods, services and related information** between the point of origin and the point of consumption (CSCMP definition). In plain words: it **moves and stores** things.

2. **What is its target?**

   The **7 Rights**: the right product, in the right quantity, in the right condition (quality), at the right place, at the right time, to the right customer, at the right cost.

3. **What does SCM do that logistics does not?**

   SCM also manages **relationships** and **decisions across firms**: choosing and developing suppliers, sourcing and procurement, demand planning, production planning, customer relationships, and the **money and information flows**.

4. **So what is the relationship?**

   Logistics is a **subset** of SCM. **SCM is the whole system; logistics is the physical engine inside it.** Without logistics the chain cannot deliver; without SCM, logistics optimises its own piece but the chain as a whole may still fail (for example, cheap trucking that causes stockouts at the retailer).

5 **Compare them side by side.**

	LOGISTICS	SCM
Scope	Movement and storage of goods	The whole network of firms and all flows
Focus	Operational, efficiency of flows	Strategic, coordination and collaboration
Main question	"How do we deliver it?"	"Who makes, buys and delivers what, and how do we coordinate?"
Activities	transport, warehousing, inventory, packaging, order fulfilment, reverse logistics	sourcing, supplier relationships, demand and production planning, customer relationships, plus logistics
Goal	7 Rights at low cost	Customer value and competitiveness of the whole chain

6 **Example.**

For an online shop, **logistics** is the warehouse picking the order, the courier delivering it and the return pickup. **SCM** additionally decides which supplier to buy from, how much to stock for the 11.11 sale, how to share sales data with suppliers and how payment terms work.

ILLUSTRATION

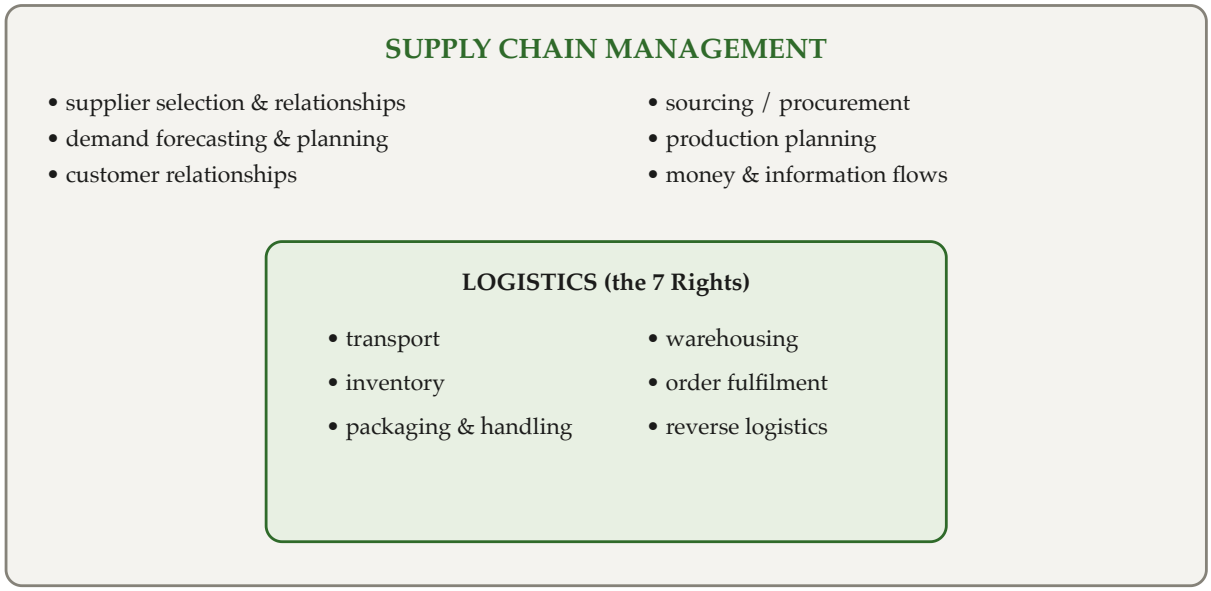


FIGURE 7.1 *Logistics (inner box, the 7 Rights) sits inside SCM (outer box), which adds relationships, planning, money and information.*

> **Draw this in the exam:** a big rectangle labelled SCM with a smaller rectangle inside labelled Logistics; list 4–5 activities in each. You can also point to the green material arrows on the SCM network diagram (Q6) and say "this part is logistics".

EXAM-READY ANSWER

MODEL ANSWER

Logistics is a part (subset) of Supply Chain Management (SCM). Logistics plans, implements and controls the efficient forward and reverse **flow and storage of goods, services and related information** from point of origin to point of consumption, aiming at the **7 Rights**: right product, quantity, condition, place, time, customer and cost. Its activities are transport, warehousing, inventory, packaging and handling, order fulfilment and reverse logistics.

SCM is broader and more strategic: it coordinates the **whole network of firms** and **all three flows** (materials, information and money), including supplier selection and relationships, sourcing and procurement, demand and production planning and customer relationships.

So SCM is the whole system and logistics is its physical engine: logistics delivers what SCM plans. They depend on each other: excellent logistics cannot fix a poorly coordinated chain, and a good SCM plan fails without reliable logistics.

Example: in e-commerce, logistics picks, ships and handles returns of an order, while SCM decides suppliers, stock levels for a sales campaign and data sharing with partners.

COMMON MISTAKES

- ✗ Treating the two as the same thing, or as completely separate.
- ✗ Forgetting that logistics includes **reverse** flows.
- ✗ Not drawing the nested diagram, which is the clearest proof of understanding.

QUESTION 8 · KISI-KISI

Order winners and order qualifiers: define them with examples

WHAT THE QUESTION REALLY ASKS

Define both terms, show the difference clearly, give examples, and ideally explain that the two change over time.

Keywords to hit: minimum standard to be considered, decisive factor to win the order, competitive priorities (cost, quality, speed, flexibility), winners become qualifiers.

THINKING IT THROUGH, STEP BY STEP

1. **Think like a customer.**

   When you buy something, you decide in **two stages**:

   1. First you **filter**: you ignore anything that fails your minimum (no safety certificate, no halal label, wrong size).
   2. Then you **choose** among the survivors based on what makes one stand out (lowest price, best camera, fastest delivery).

2. **Name each stage.**

   - **Order qualifiers** are the characteristics a product or service **must have just to be considered**. Meeting them does not win the sale; failing them **loses** it.
   - **Order winners** are the characteristics that **make the customer choose you over competitors**. Being better on them wins the order.

3. **Why does this matter for operations?**

   Operations cannot be the best at everything. The firm should **meet the qualifiers adequately** (overspending there is waste) and **excel at the winners**. This is how strategy turns into operations decisions about cost, quality, speed and flexibility.

4. **Examples.**

MARKET	ORDER QUALIFIERS (MUST HAVE)	ORDER WINNERS (MAKES YOU CHOOSE)
Low-cost airline (e.g. Citilink, AirAsia)	safety, reasonable on-time record, the route exists	**lowest fare**

MARKET	ORDER QUALIFIERS (MUST HAVE)	ORDER WINNERS (MAKES YOU CHOOSE)
Full-service airline (e.g. Garuda)	safety, the route exists	service quality, comfort, frequent-flyer benefits, network connections
Smartphone	4G/5G, reliable battery, safety certification, app store	camera quality, ecosystem, brand, price for the specs
Industrial supplier (from the slides: supplier selection)	capability, quality certifications, delivery reliability, conformance to specifications, financial capability, ethical requirements	the lowest price or cost reduction, fastest lead time, responsiveness
Fast food	hygiene, halal, acceptable taste	speed, price, location, promotions

5 **They change over time (an important exam point).**

Today's winner becomes tomorrow's qualifier once every competitor copies it. Examples: quality in cars in the 1980s; free Wi-Fi in hotels; online tracking for couriers. The slides put it as: **"order winners have become order qualifiers"**, so firms now have to compete through their **whole supply chain** to find new winners.

6 **A related term.**

An **order loser** is a qualifier that drops below the minimum, e.g. a single late delivery that makes a buyer drop a supplier.

ILLUSTRATION

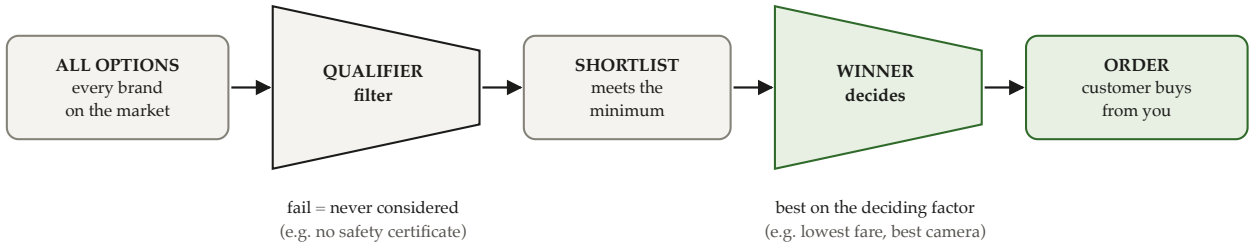


FIGURE 8.1 *All options pass through the qualifier filter first (fail = never considered). Among the shortlist, the order winner decides who gets the order.*

> **Draw this in the exam:** a funnel or two filters in a row: "All options → Qualifier filter → Shortlist → Winner → Order", with one example written under each filter.

EXAM-READY ANSWER

MODEL ANSWER

Order qualifiers are the characteristics a product or service must have **to be considered** by customers at all; meeting them does not win the sale, but failing them loses it. **Order winners** are the characteristics that make customers **choose one firm over its competitors**. Customers first filter options with qualifiers, then choose with winners.

Example: for a low-cost airline, safety and a reasonable on-time record are qualifiers, while the lowest fare is the order winner. For a smartphone, 5G and a safe, reliable battery are qualifiers, while camera quality or the ecosystem may win the order.

This matters for operations strategy: a firm should meet qualifiers adequately and focus resources on excelling at the winners. Winners also **become qualifiers over time** as competitors copy them (e.g. quality, free Wi-Fi, parcel tracking), which is why firms now compete through their whole supply chain to create new winners.

COMMON MISTAKES

- ✗ Saying qualifiers are "less important". They are essential, they simply do not differentiate.
- ✗ Giving examples without saying *which customer segment*. The same factor can be a winner for one segment and a qualifier for another.

QUESTION 9 · KISI-KISI

Point-to-point vs hub-and-spoke: illustrate both and decide which is better

WHAT THE QUESTION REALLY ASKS

Explain both network designs (from the airline case), draw them, compare advantages and disadvantages, and give a reasoned verdict. "Which is better?" is a trap: the strong answer is **"it depends on…"** followed by clear conditions.

Keywords to hit: direct routes, hub, transfer, number of routes, load factor, economies of scale, coverage, travel time, congestion and delays, full-service vs low-cost carrier.

THINKING IT THROUGH, STEP BY STEP

1. **State the problem.**

   An airline (or courier, or distributor) must connect many cities. Should it fly **directly between every pair**, or bring everything **through one central point**?

2. **Define both.**

   - **Point-to-point (P2P):** direct routes between cities. Passengers or goods travel straight from origin to destination.
   - **Hub-and-spoke:** every city (spoke) connects to a central **hub**; trips between two spokes **transfer at the hub**.

3. **Count the routes (this is the core logic).**

   With **n** cities:

   - Point-to-point needs up to **n(n − 1) / 2** routes to connect every pair. For 6 cities: 6 × 5 / 2 = **15 routes**.
   - Hub-and-spoke needs only **n − 1** routes. For 6 cities: **5 routes**, yet every pair is still connected (with one transfer).

   So the hub design **covers the same network with far fewer routes**, which means planes fly fuller and the cost per passenger falls (economies of scale and higher load factors).

4. **But what does the hub cost?**

   Longer total travel time (detours and a connection), the risk of missed connections, **congestion at the hub**, and **delays that spread** through the whole

network when the hub has a problem (weather, strike).

5 **Compare.**

	POINT-TO-POINT	HUB-AND-SPOKE
Routes needed (6 cities)	15	5
Travel time	short, direct	longer, needs a transfer
Load factor on thin routes	low (half-empty planes)	high (traffic pooled at the hub)
Network coverage	limited to busy pairs	many destinations reachable
Cost structure	simple, fast turnaround, fewer staff	expensive hub operations, more coordination
Risk	a delay stays local	a hub disruption spreads everywhere
Typical user	**low-cost carriers** (Southwest Airlines, many LCCs)	**full-service carriers** (American Airlines at Dallas/Fort Worth, Garuda at Jakarta)

6 **Decide with conditions (the verdict).**

- **Choose point-to-point** when demand between two cities is **high enough to fill planes directly** (dense routes), when **speed and low cost** are the order winners, and when the business model is simple, like a low-cost carrier. Example: Jakarta–Surabaya, Jakarta–Denpasar.
- **Choose hub-and-spoke** when there are **many thin routes** (small demand between each pair), when **wide coverage** matters, and when the carrier sells connections (full-service, international transfers). Example: connecting Medan, Balikpapan, Makassar, Denpasar and Surabaya to each other through Jakarta.
- **In practice most firms use a hybrid**: hubs for the thin routes and direct flights on the busiest pairs. Couriers (e.g. FedEx at Memphis) and parcel networks use hubs for sorting too.

7 **Link to supply chains.**

The same choice appears in distribution: ship **direct from factory to store** (point-to-point) or **through a distribution centre or cross-dock** (hub). Walmart's cross-docking is a hub idea.

ILLUSTRATION

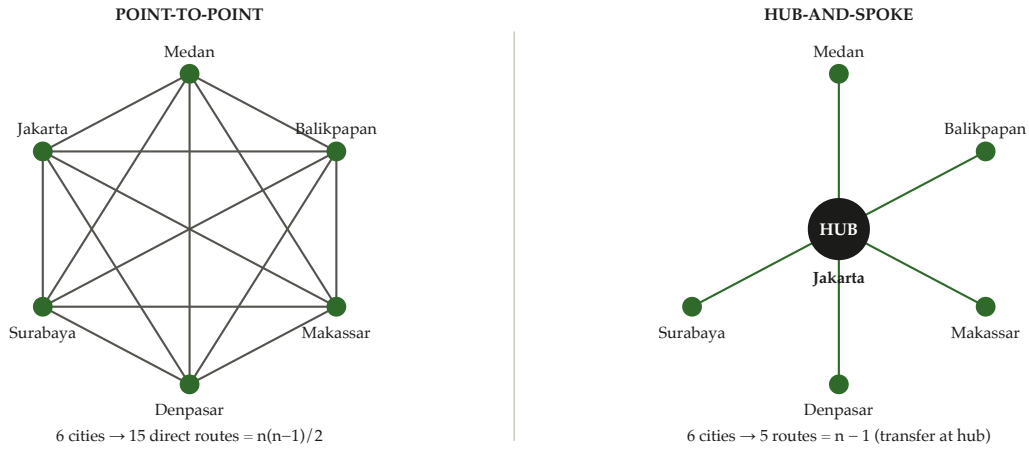


FIGURE 9.1 *Six Indonesian cities. Point-to-point: n(n−1)/2 = 15 direct routes. Hub-and-spoke through Jakarta: n − 1 = 5 routes, with transfers at the hub.*

Draw this in the exam: two hexagons of six dots. Left: join every dot to every other (15 lines). Right: one big central dot (HUB) joined to each outer dot (5 lines). Write the two formulas underneath.

EXAM-READY ANSWER

MODEL ANSWER

In a **point-to-point** network every city pair is linked by direct routes, while in a **hub-and-spoke** network all cities (spokes) connect through a central **hub** where passengers or goods transfer. For n cities, point-to-point needs up to n(n−1)/2 routes (15 for 6 cities), while hub-and-spoke needs only n−1 (5 for 6 cities).

Hub-and-spoke pools traffic at the hub, giving fuller planes, economies of scale and wide coverage with fewer routes; but trips are longer, need transfers, and congestion or disruption at the hub spreads through the whole network. **Point-to-point** gives faster, direct trips, simpler operations and quicker aircraft turnaround, and a delay stays local; but it only pays on routes dense enough to fill planes, and coverage is limited.

Which is better depends on demand density and strategy. For a **low-cost carrier** competing on price and speed on busy routes (e.g. Southwest Airlines, or Jakarta–Surabaya), point-to-point is better. For a **full-service carrier** serving many thinner routes and selling connections (e.g. American Airlines at

Dallas/Fort Worth, Garuda via Jakarta), hub-and-spoke is better. Most firms use a **hybrid**: direct flights on dense routes and a hub for the rest. The same logic applies to distribution: direct shipment vs a distribution centre or cross-dock.

COMMON MISTAKES

✗ Answering "hub-and-spoke is better" (or the reverse) without conditions.

✗ Not quantifying the route difference. The formula shows real understanding.

✗ Forgetting the downside of hubs: one failure point that spreads delays.

QUESTION 10 · KISI-KISI

Lean operations: define Lean and explain the whole concept

WHAT THE QUESTION REALLY ASKS

Explain where lean comes from, what it means, its principles, the wastes, its tools (JIT, jidoka, kanban, kaizen), and how it changes operations and supply chains.

Keywords to hit: Toyota Production System (TPS), value from the customer's view, waste (muda), 5 principles, 8 wastes, push vs pull, just-in-time (JIT), jidoka, heijunka, kaizen, respect for people.

THINKING IT THROUGH, STEP BY STEP

1. **What was wrong before lean?**

   Production evolved from **craft** (custom, one by one) to **mass production** (Henry Ford: standard products, long runs, low cost per unit). Mass production is a **push** system: produce to a forecast in big batches, store the result (make-to-stock). When the forecast is wrong, you get **huge stocks, hidden defects and slow change**.

2. **Toyota's answer.**

   After the Second World War, Toyota had little money or space for big stocks. Taiichi Ohno and Eiji Toyoda built the **Toyota Production System (TPS)**: make only what is needed, when it is needed, in the amount needed. The West later called this **lean** (it uses *less* of everything: inventory, human effort, space, tooling, engineering hours).

3. **Definition.**

   Lean operations is a way of managing that **maximises customer value by systematically eliminating waste** and continuously improving the flow of products and information, producing **only on real demand (pull)**.

4. **What counts as waste? Anything the customer would not pay for (Japanese: *muda*).**

   The 8 wastes, mnemonic **TIM WOODS**:

WASTE	SUPPLY CHAIN EXAMPLE
Transport	double handling between warehouses
Inventory	excess safety stock
Motion	warehouse pickers walking long routes
Waiting	trucks queuing at docks
Overproduction	making before orders arrive (the worst one, it hides the others)
Over-processing	extra packaging, unnecessary checks
Defects	damaged goods, returns, rework
Skills (8th)	not using people's ideas and abilities

5 **How to remove waste: the 5 lean principles (in order).**

1. **Value**: define value from the customer's point of view.
2. **Value stream**: map every step that creates it; mark the steps that add no value.
3. **Flow**: make the remaining steps flow without waiting or batching.
4. **Pull**: produce only when the customer (or the next process) pulls.
5. **Perfection**: find root causes and keep improving forever.

6 **Push vs pull (the heart of lean).**

- **Push:** forecast → make in big batches → store → hope it sells (make-to-stock, MTS).
- **Pull:** a real order triggers production → materials arrive just in time → small lots → no big buffer (make-to-order, MTO). The signal is often a **kanban** card: an empty bin tells the previous step to make more.

7 **The tools, arranged as the "lean house".**

- **Roof (the goal):** highest quality, lowest cost, shortest lead time.
- **Pillar 1, Just-In-Time (JIT):** pull, takt time (produce at the pace of demand), continuous flow, kanban.
- **Pillar 2, Jidoka** (automation with a human touch): stop the line when a defect appears, build quality in, error-proofing (poka-yoke). It began on Toyoda's automatic looms.
- **Centre:** waste elimination and **respect for people**, teamwork.

- **Foundation: heijunka** (level production, no surges or idle time), **standardised work**, **kaizen** (small continuous improvement by everyone), and stability of machines, materials and people.

8 **Results, and when lean fits.**

Lower cost and inventory, better quality, faster response, and problems become visible quickly. Lean works best with **stable, predictable demand** and reliable suppliers (functional products in Fisher's model). Its risk: very little buffer, so a disruption (earthquake, pandemic, chip shortage) can stop the line. The answer is lean plus risk management, not abandoning lean.

9 **Lean in the supply chain.**

Lean extends beyond the factory: **vendor-managed inventory (VMI)**, cross-docking, small frequent deliveries, sharing POS data with suppliers, and **lean consumption** (solve the customer's problem completely without wasting their time).

ILLUSTRATION

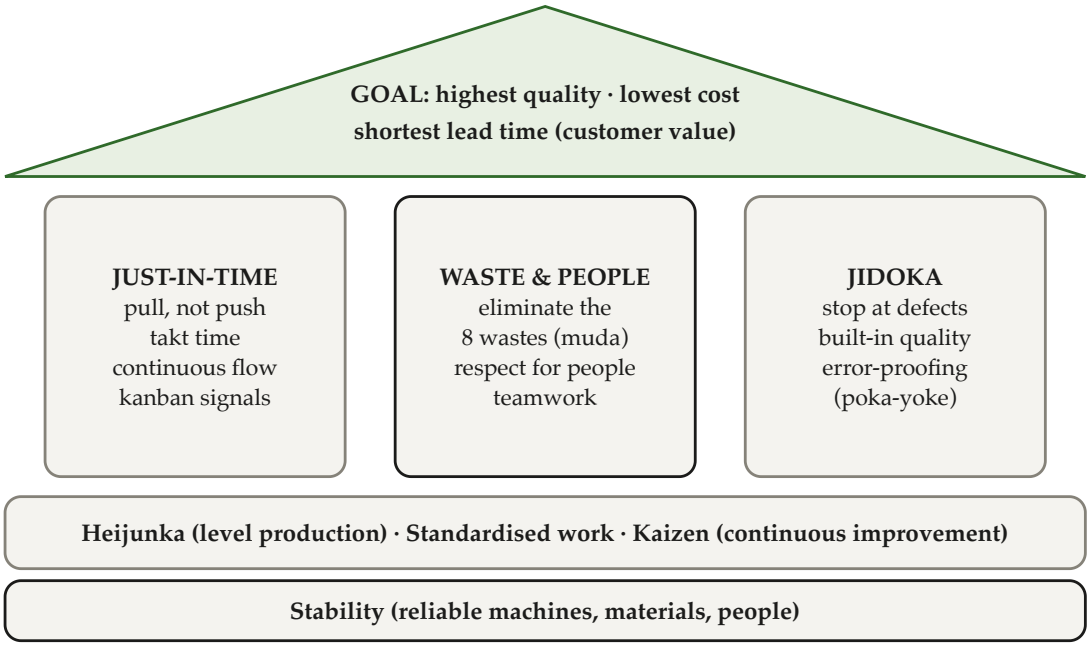


FIGURE 10.1 *The lean house: two pillars (JIT and jidoka) hold up the goal; people and waste removal sit in the middle; stability, level production and kaizen are the foundation.*

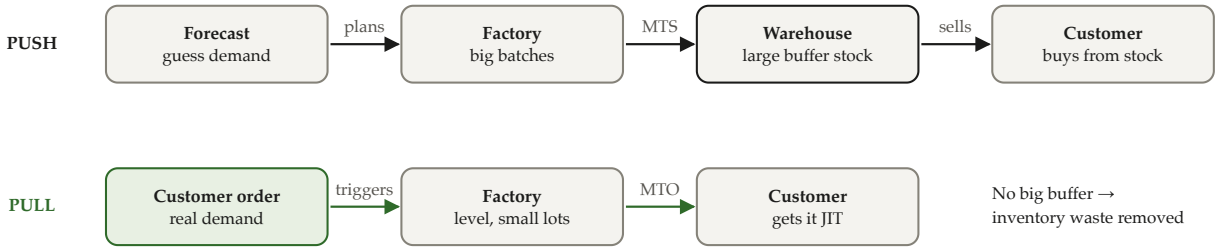


FIGURE 10.2 *Push builds to a forecast and stores the result. Pull starts only when real demand arrives, so the big buffer stock disappears.*

Draw this in the exam: the lean house (triangle roof, two pillars, a base) and, beside it, two lines comparing push and pull. Write TIM WOODS down the side.

EXAM-READY ANSWER

MODEL ANSWER

Lean operations is a management approach that **maximises customer value by eliminating waste** and producing only according to real demand. It comes from the **Toyota Production System (TPS)** developed by Taiichi Ohno, as an alternative to mass production, which pushed products to a forecast and created large inventories and hidden defects.

Waste (muda) is anything the customer would not pay for. The 8 wastes are transport, inventory, motion, waiting, overproduction, over-processing, defects and unused skills (TIM WOODS).

Lean follows **five principles**: define **value** from the customer's view, map the **value stream**, create **flow**, let the customer **pull**, and pursue **perfection**.

Instead of **push** (make-to-stock from a forecast), lean uses **pull**: production starts when real demand arrives, with materials arriving **just-in-time (JIT)**, often signalled by **kanban**. The "lean house" has two pillars, **JIT** and **jidoka** (stop and fix defects immediately, build quality in), on a foundation of **heijunka** (level production), standardised work and **kaizen** (continuous small improvements by everyone), with respect for people at its centre.

The results are lower cost and inventory, higher quality and faster response. Lean fits stable demand best; because it holds little buffer, it must be paired

with supply chain risk management. In supply chains it appears as VMI, cross-docking, small frequent deliveries and shared demand data.

COMMON MISTAKES

- ✗ "Lean = cutting staff or cutting costs." Lean cuts **waste**, and respect for people is one of its pillars.
- ✗ Listing the wastes without examples.
- ✗ Forgetting push vs pull, which is the core shift.
- ✗ Saying JIT means zero inventory. It means the **minimum needed**, delivered when needed.

QUESTION 11 · KISI-KISI

Vertical integration: forward and backward, with examples

WHAT THE QUESTION REALLY ASKS

Explain what vertical integration is, the two directions, real examples of each, and why firms do it (or avoid it).

Keywords to hit: ownership of more stages of the supply chain, make vs buy, backward (towards suppliers), forward (towards customers), control, cost, risk, flexibility.

THINKING IT THROUGH, STEP BY STEP

1. **The underlying decision: make or buy?**

   For each stage of the chain (raw material, components, distribution, retail), a firm can **buy** it from another company or **own and do it** itself. Owning more stages is vertical integration.

2. **Definition.**

   Vertical integration is the degree to which a firm **owns or controls several stages of its supply chain**, upstream or downstream of its core business. (It is different from *horizontal* integration, which means buying competitors at the same stage.)

3. **The two directions.**

   Stand at the **focal firm** and look along the chain:

   - **Backward (upstream) integration**: owning **suppliers**, the stages that feed you (raw materials, components).
   - **Forward (downstream) integration**: owning **distribution and retail**, the stages that reach the customer.

4. **Examples.**

DIRECTION	COMPANY	WHAT IT OWNS
Backward	**Indofood** (instant noodle maker)	**Bogasari** flour mills that supply its wheat flour

DIRECTION	COMPANY	WHAT IT OWNS
Backward	**Tesla**	produces its own battery cells and much of its software
Backward	**Starbucks**	its own coffee farm (in Costa Rica) for research and supply
Backward	**Netflix**	produces its own films and series instead of only licensing them
Forward	**Apple**	Apple Stores and its online store, selling directly to consumers
Forward	**Nike**	Nike stores and Nike app (direct-to-consumer)
Both	**Zara (Inditex)**	its own design, many of its factories, logistics and stores, which enables its fast supply chain

5 **Why integrate? (benefits)**

- **Control** over quality, supply security and timing (no supplier hold-up).
- **Lower cost** by removing middlemen's margins and transaction costs.
- **Better coordination and information**, so less bullwhip and faster response (Zara).
- **Protect secrets** and capabilities; own the customer relationship (forward).

6 **Why not? (risks)**

- **High investment** and fixed costs.
- **Less flexibility**: you are stuck with your own supplier even when better or cheaper ones appear.
- **Loss of focus** on the core competence; the new stage may need skills you do not have.
- **Weaker incentives**: an internal supplier faces no competition.

7 **So what is the decision rule?**

Integrate when the stage is **strategic** (critical to quality or supply, or a core capability), when supply is **risky or scarce**, or when it creates a real cost advantage. **Outsource** when the market for that stage is competitive and reliable and it is not a core competence. This links to 1PL–5PL (Q6): outsourcing logistics to a 3PL is the opposite of integrating it.

ILLUSTRATION

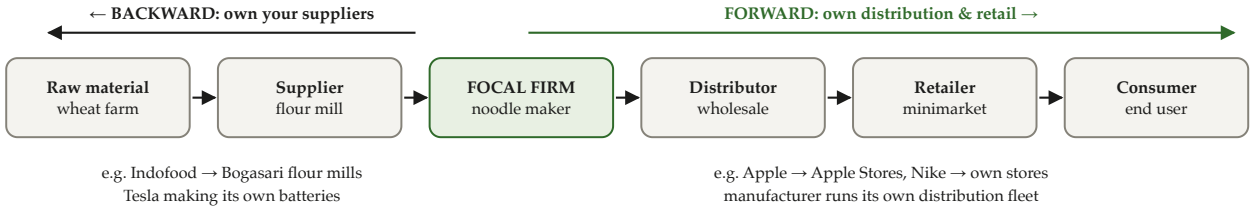


FIGURE 11.1 *From the focal firm (a noodle maker), backward integration owns the flour mill and farm; forward integration owns the distributor or retailer.*

Draw this in the exam: a row of boxes Raw material → Supplier → **Focal firm** → Distributor → Retailer → Consumer; an arrow pointing left labelled "backward" and one pointing right labelled "forward", with one example under each.

EXAM-READY ANSWER

MODEL ANSWER

Vertical integration is the extent to which a firm **owns or controls multiple stages of its supply chain** instead of buying them from other companies; it is a make-or-buy decision. **Backward (upstream) integration** means owning suppliers of inputs, e.g. **Indofood** owning **Bogasari** flour mills for its noodles, or **Tesla** making its own battery cells. **Forward (downstream) integration** means owning distribution or retail closer to the customer, e.g. **Apple** Stores or **Nike** direct-to-consumer stores. **Zara** integrates in both directions, which lets it move from design to store in weeks.

Benefits: more control over quality and supply, lower costs by removing intermediaries' margins, better coordination and information sharing (less bullwhip), protection of know-how and closer customer relationships. **Drawbacks:** high investment, less flexibility to switch to better suppliers, loss of focus on core competences and weaker cost discipline without market competition.

Firms should integrate stages that are strategic, risky or a source of cost advantage, and outsource stages where competitive, reliable markets already exist.

COMMON MISTAKES

- ✗ Mixing up the directions. **Backward = towards suppliers. Forward = towards customers.**
- ✗ Confusing vertical integration with horizontal integration (buying a competitor).
- ✗ Giving only benefits. A good answer weighs the drawbacks.

QUESTION 12 · KISI-KISI

The technology drivers of the industrial revolutions (steam engine, electric light, Ford, …)

WHAT THE QUESTION REALLY ASKS

Name the key technologies that drove each industrial revolution, explain how each changed production and operations, and connect them to today. The kisi-kisi mentions the steam engine (1.0), Ford and the electric lamp (2.0), so cover all four revolutions.

Keywords to hit: mechanisation, steam power, electricity, mass production, assembly line, scientific management, automation, computers, cyber-physical systems, Industry 4.0.

THINKING IT THROUGH, STEP BY STEP

1. **The logic of every revolution.**

   A **new source of power or control** appears → it changes **how work is organised** → output and productivity jump → supply chains grow wider. Use this chain for each era.

2. **Industry 1.0 (about 1760–1840): mechanisation with water and steam.**
   - **Technology:** the **steam engine** (improved by **James Watt**, patented 1769), mechanised textile machines (spinning jenny, power loom), later railways and steamships.
   - **Ideas: division of labour** (Adam Smith, *The Wealth of Nations*, 1776); **interchangeable parts** (Eli Whitney, around 1800).
   - **What changed:** work moved from homes (craft) to **factories**; machines replaced muscle power; railways let goods travel further, so markets and supply chains widened.

3. **Industry 2.0 (about 1870–1914): electricity and mass production.**
   - **Technology: electricity** and the electric motor; the practical **light bulb** (**Thomas Edison**, 1879), which allowed factories to work longer shifts and indoors at night; steel, the telephone, oil and the internal-combustion engine.
   - **Ideas: scientific management** (**Frederick Taylor**, 1911): study each task, set standard methods and times; the **Gantt chart** for scheduling.

- **Ford: Henry Ford's moving assembly line** (1913) for the Model T cut the assembly time of a car from about 12 hours to about 1.5 hours and made cars affordable. This is **mass production**: standard products, economies of scale, low cost per unit.
- **What changed:** huge volumes at low cost; OM became a discipline of standard work and efficiency.

4 **Industry 3.0 (about 1960s–1990s): electronics and information technology.**

- **Technology:** computers, **programmable logic controllers (PLC)** that automate machines, robots, the internet; software such as **MRP** (Material Requirements Planning) and **ERP** (Enterprise Resource Planning).
- **What changed: automation** and data; lean/TPS and quality management spread; **global supply chains** became possible because firms could coordinate suppliers worldwide.

5 **Industry 4.0 (2010s–now): cyber-physical systems.**

- **Technology:** the **Internet of Things (IoT)**, artificial intelligence and big data, cloud computing, advanced robots, 3D printing, **digital twins**.
- **What changed:** machines and products talk to each other; real-time visibility across the chain; mass customisation; predictive maintenance; smarter, more sustainable supply chains. (Some now speak of **Industry 5.0**: human-centric, resilient, sustainable.)

6 **The pattern to state in the conclusion.**

Each revolution **raised productivity** and **lengthened supply chains**: from local workshops (craft) → national factories → global networks → digitally connected networks. That is why supply chain management became necessary.

ILLUSTRATION

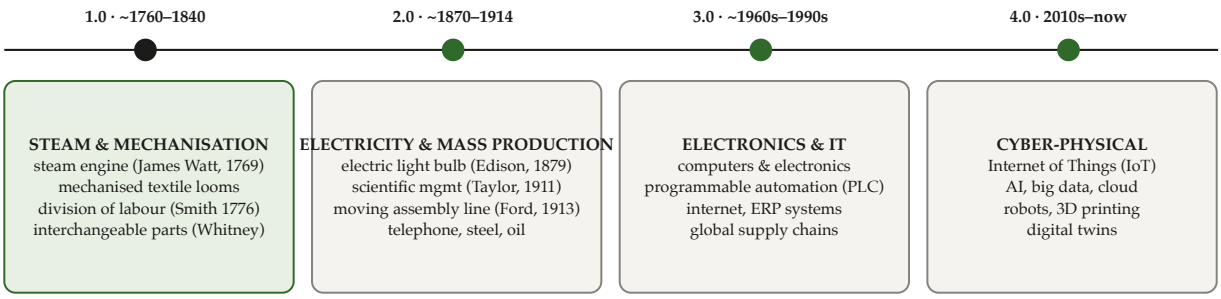


FIGURE 12.1 *Four revolutions, four new drivers: steam, electricity and mass production, electronics and IT, cyber-physical systems.*

Draw this in the exam: a horizontal timeline with four dots: 1.0 Steam (Watt 1769) · 2.0 Electricity, bulb (Edison 1879), assembly line (Ford 1913) · 3.0 Computers/PLC/internet · 4.0 IoT/AI/cloud.

EXAM-READY ANSWER

MODEL ANSWER

Each industrial revolution was driven by a new technology that changed how production is organised:

Industry 1.0 (from about 1760): mechanisation. The **steam engine**, improved by **James Watt** (1769), powered textile machines and later railways and ships. Production moved from home crafts to factories, supported by Adam Smith's **division of labour** (1776) and Eli Whitney's **interchangeable parts**.

Industry 2.0 (from about 1870): electricity and mass production. Electricity and **Edison's light bulb** (1879) let factories run machines with electric motors and work longer hours. **Frederick Taylor's scientific management** (1911) standardised work, and **Henry Ford's moving assembly line** (1913) made the Model T quickly and cheaply, the start of **mass production**.

Industry 3.0 (from the 1960s–70s): electronics and IT. Computers, **programmable logic controllers** and robots automated production, and IT systems (MRP, ERP, the internet) made global supply chains manageable.

Industry 4.0 (2010s–now): cyber-physical systems. The **Internet of Things**, AI and big data, cloud computing, robotics, 3D printing and digital twins connect

machines and partners in real time, enabling mass customisation and smart supply chains.

Each revolution increased productivity and widened supply chains, from local workshops to connected global networks.

COMMON MISTAKES

- ✗ Putting Ford in the first revolution. **Ford's assembly line (1913) belongs to Industry 2.0.**
- ✗ Saying James Watt "invented" the steam engine. He greatly **improved** it (Newcomen's engine came first, 1712).
- ✗ Listing technologies without saying how each changed operations.

QUESTION 13 · KISI-KISI

ESG, competition, the social pillar, stockholders vs stakeholders, CSR and 3R

WHAT THE QUESTION REALLY ASKS

Explain the sustainability vocabulary from Sessions 9–10 and how the ideas connect. The lecturer **flagged this topic** with specific examples, so use them: governance (coaching vs dismissing), Apple–Foxconn (social), Samsung–Apple (competition), CSR as vision and mission vs ESG as targets.

Keywords to hit: 3P (People, Planet, Profit), Environmental, Social, Governance, fair labour, human rights, stockholder, stakeholder, CSR, coopetition, promoting scale, Reduce/Reuse/Recycle, recovery hierarchy.

THINKING IT THROUGH, STEP BY STEP

1 **Why is this in a supply chain course?**

A firm's impact does not stop at its own walls: most of its emissions, labour practices and waste sit in its **supply chain** (suppliers' factories, transport, packaging, product disposal). Customers, investors and governments now hold the brand responsible for all of it. So sustainability is a supply chain decision.

2 **The big idea: 3P, the triple bottom line.**

Why it exists: traditionally a company was judged on one line only, the **bottom line** of the income statement (profit). But a firm can make a profit while hurting its workers or polluting a river, and those costs are paid by someone else. So John Elkington (1994) proposed judging a business on **three bottom lines** instead of one. The lecturer calls it a **management concept**: a way to decide and to evaluate.

What each P means, and what it looks like in a supply chain:

P	QUESTION IT ASKS	IN THE SUPPLY CHAIN	EXAMPLE
People (social)	Are the humans touched by our business treated fairly?	fair wages and hours, safe factories, no child or forced labour **at suppliers too**, fair treatment of customers, support for local communities	Apple had to audit and improve working conditions at its supplier Foxconn
Planet (environmental)	Are we protecting the environment we depend on?	lower emissions from transport, less packaging, energy and water saving, recycling, reverse logistics	returnable AQUA gallons; fuller trucks through consolidated loads
Profit (economic)	Do we create economic value that lasts?	a fair return for owners, jobs, taxes, fair prices and prompt payment to suppliers, growth of local small suppliers (UMKM)	a firm that loses money cannot pay workers or invest in greener trucks

The key reasoning: the three are **balanced, not traded off one by one**. Profit is not the enemy: without profit the firm dies and helps nobody. But profit earned by hurting people or the planet is not **sustainable**, because it creates risks (boycotts, strikes, fines, lost licences, scarce resources) that eventually destroy the profit too. A sustainable business sits in the **middle of the triangle**, where all three are satisfied.

How 3P connects to ESG and CSR (this is the logic to show in the exam):

- **CSR = WHY**: the company's ethical foundation, values and vision ("we commit to treat people fairly and protect the planet while staying profitable").
- **3P = WHAT**: the three things the company commits to balance.
- **ESG = HOW it is measured**: each P becomes measurable targets that are reported. **People → S (Social)**, **Planet → E (Environmental)**, and **Profit → G (Governance)**, because good governance (ethical leadership, compliance, accountability) is what keeps profit honest and long-term, and investors read ESG reports to judge long-term risk to their returns.

3 **ESG turns 3P into measurable areas: Environmental, Social and Governance.**

- **Environmental:** emissions, energy, water, waste, packaging.

- **Social: fair labour practice, community engagement and human rights**, including in suppliers' factories. *Example (lecturer):* **Apple and its Chinese supplier Foxconn**: after worker suicides and harsh conditions at Foxconn around 2010, Apple was held responsible and had to audit and improve supplier working conditions. The lesson: you are judged by your suppliers' behaviour.
- **Governance: how the company is run**: ethics, transparency, board oversight, anti-corruption, fair treatment of employees. *Example (lecturer):* when an employee performs badly, good governance means **coaching first**; if it must end, **offer the choice to resign** rather than dismissing abruptly.

4 **Who is the firm responsible to? Stockholders vs stakeholders.**

- **Stockholders (shareholders):** the people who **own shares**. The traditional view (stockholder theory) says the firm's job is to maximise their return.
- **Stakeholders: everyone affected** by the firm: employees, customers, suppliers, the community, government, the environment, and the stockholders too. Stakeholder theory says the firm must balance all of them.
- The relationship: **stockholders are a subset of stakeholders.** ESG and 3P follow the stakeholder view.

5 **CSR vs ESG (how they differ).**

- **CSR (Corporate Social Responsibility):** the firm's voluntary commitment to act responsibly. In the lecturer's words, CSR gives the **vision and mission**: the *why* and the intention (e.g. community programmes, donations, volunteering).
- **ESG:** sets **measurable targets that must be achieved and reported** (e.g. cut emissions 30% by 2030, 100% of suppliers audited). Investors use ESG scores to judge risk.
- In one line: **CSR = commitment (what we believe); ESG = measurement (what we achieve).**

6 **Competition can also be cooperation (coopetition).**

Firms can **compete and cooperate at the same time**. *Example (lecturer):* **Samsung competes with Apple** in smartphones, yet **Samsung Display supplies OLED screens for iPhones**. Cooperation can also be sustainable:

competitors sharing trucks or warehouses to **promote scale**, i.e. consolidating loads so fewer, fuller trucks drive, which cuts both cost and emissions.

7 **3R: Reduce, Reuse, Recycle, and the recovery hierarchy.**

- **Reduce:** use less material and energy in the first place (best).
- **Reuse:** use the product again as it is (returnable bottles, pallets).
- **Recycle:** turn the material into new raw material (plastic, aluminium).

The slide's **recovery hierarchy** (from most to least value kept): **resource reduction → reuse → remanufacturing → recycling → disposal with energy recovery → landfill**. Always choose the highest option possible. Collecting products back for these options is **reverse logistics** (Q14).

8 **How it all connects.**

CSR states the commitment (why) → **3P** names what to balance: people, planet, profit (what) → **ESG** turns each P into measurable targets: Social, Environmental, Governance (how it is measured) → all of it serves **stakeholders**, not only stockholders → **coopetition** and **promoting scale** make it efficient → **3R** and **reverse logistics** deliver the planet part inside the supply chain.

ILLUSTRATION

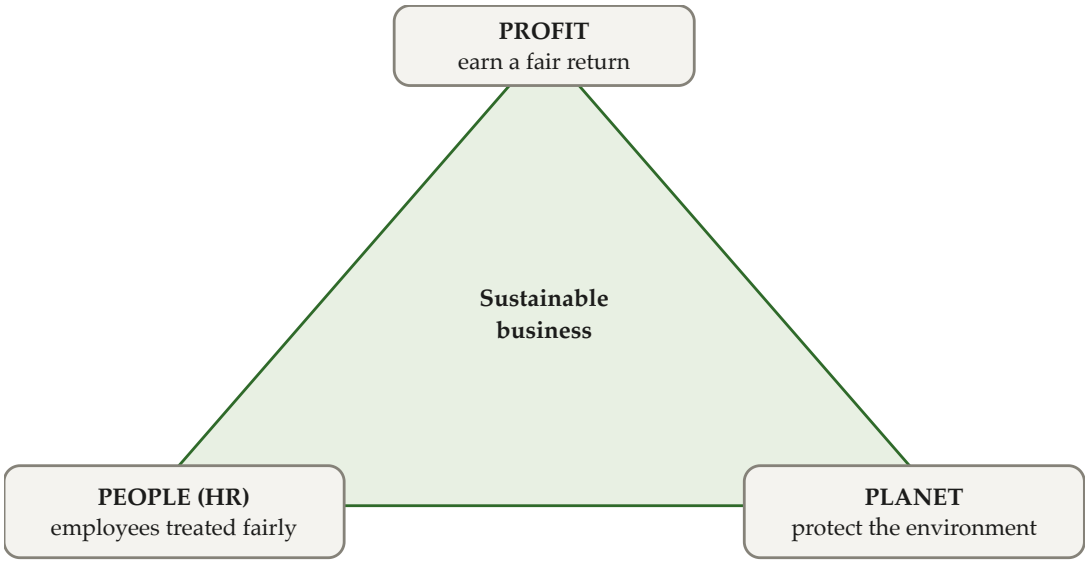


FIGURE 13.1 *A sustainable business balances all three corners.*

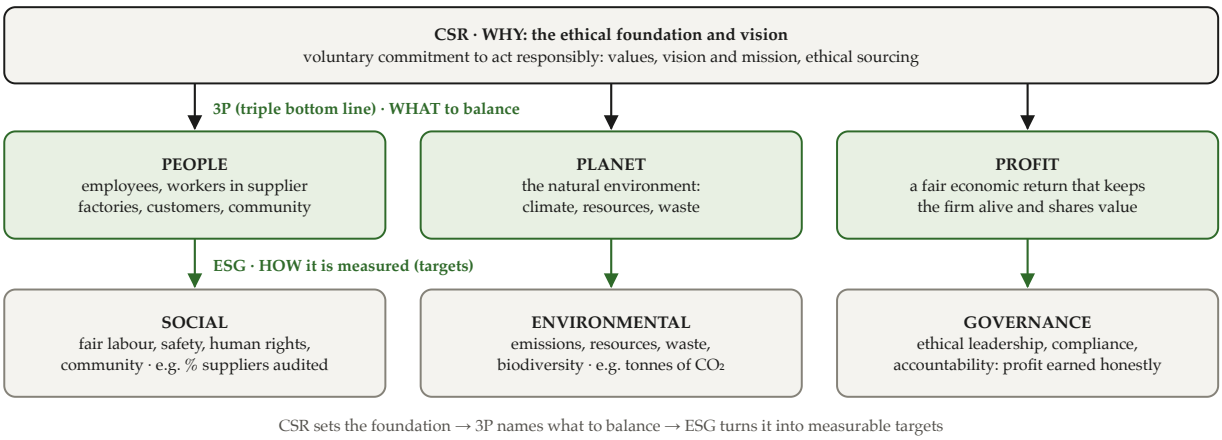


FIGURE 13.2 *CSR is the why (foundation and vision), 3P is the what (people, planet, profit), ESG is the how (Social, Environmental and Governance targets). Each P feeds one ESG pillar.*

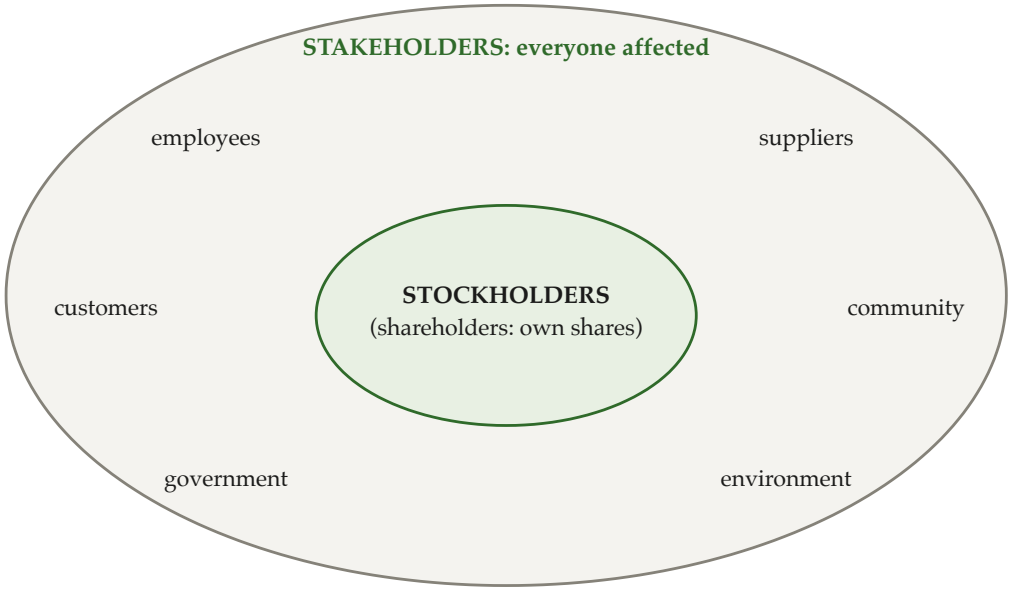


FIGURE 13.3 *Stockholders own shares; stakeholders include everyone affected. Stockholders are one group inside the stakeholders.*

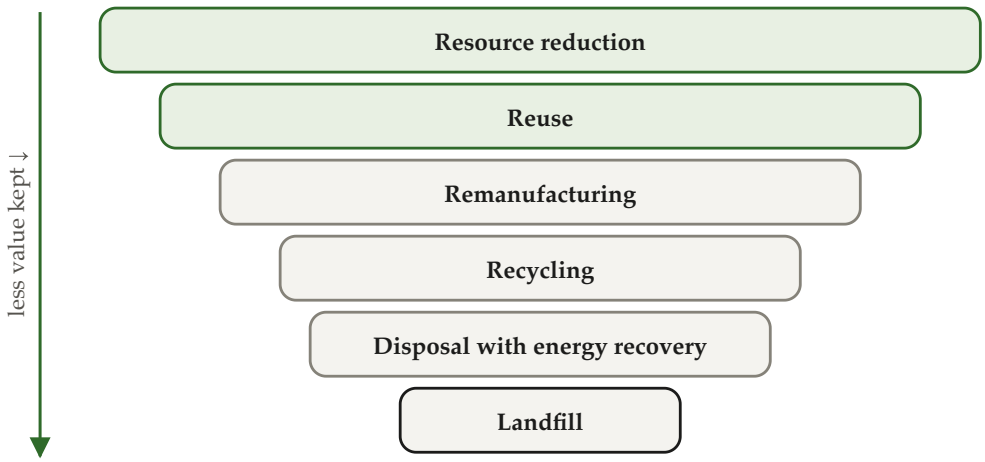


FIGURE 13.4 *3R in order of preference: the higher the option, the more value is kept.*

Draw this in the exam: the 3P triangle; under it, three arrows from People → Social, Planet → Environmental, Profit → Governance, with "CSR = why" written above; then two circles (a small "stockholders" circle inside a big "stakeholders" circle), then the recovery hierarchy as a funnel.

EXAM-READY ANSWER

MODEL ANSWER

3P (People, Planet, Profit), the triple bottom line, judges a business on three results instead of profit alone. **People** means treating employees, supplier workers, customers and communities fairly; **Planet** means protecting the environment (emissions, resources, waste); **Profit** means earning a fair, lasting economic return that keeps the firm alive and shares value with suppliers and society. They must be balanced: profit earned by harming people or the planet is not sustainable, because it brings boycotts, fines and resource risks. 3P links CSR and ESG: **CSR** is the commitment (why), **3P** is what to balance, and **ESG** measures it, with People reflected in Social, Planet in Environmental and Profit kept honest and long-term by Governance.

ESG (Environmental, Social, Governance) turns this into measurable areas. *Environmental* covers emissions, energy and waste. *Social* covers **fair labour practice, community engagement and human rights**, including in the supply chain: Apple was held responsible for poor working conditions at its supplier **Foxconn** and had to audit and improve them. *Governance* is **how the company is run**: ethics, transparency and fair treatment, e.g. coaching an under-

performing employee first and offering the option to resign rather than dismissing them abruptly.

Stockholders are the owners of shares; **stakeholders** are everyone affected by the firm (employees, customers, suppliers, community, government and stockholders). ESG follows the stakeholder view.

CSR (Corporate Social Responsibility) is a company's voluntary commitment to operate responsibly through ethical sourcing, fair labour and community engagement, across its suppliers and subcontractors. As the slides put it, **CSR sets the ethical foundation and vision, while ESG provides the measurable targets and accountability** needed to implement it.

Competition can include cooperation: **Samsung** competes with **Apple** in phones but supplies **iPhone screens**. Firms can also cooperate to **promote scale**, consolidating loads to cut cost and emissions.

3R (Reduce, Reuse, Recycle) reduces waste, following the **recovery hierarchy**: resource reduction, reuse, remanufacturing, recycling, disposal with energy recovery and, last, landfill. Returning products for these options is done by **reverse logistics**.

COMMON MISTAKES

- ✗ Using "stockholder" and "stakeholder" as synonyms.
- ✗ Saying CSR and ESG are the same. CSR = commitment; ESG = measurable targets.
- ✗ Listing People, Planet, Profit without explaining each one or how they map to ESG (People → S, Planet → E, Profit → G).
- ✗ Treating profit as "the bad P". All three are needed; the point is balance.
- ✗ Writing 3R without the order of preference (reduce first, landfill last).
- ✗ Forgetting the lecturer's examples (Foxconn, Samsung–Apple, coaching vs dismissing).

QUESTION 14 · KISI-KISI

Reverse logistics in detail

WHAT THE QUESTION REALLY ASKS

Explain reverse logistics fully: definition, why it exists, what flows back, the process steps, the recovery options, the difficulties and real examples.

Keywords to hit: point of consumption → point of origin, recapture value or proper disposal, returns, end-of-life, gatekeeping, collection, sorting/inspection, disposition, recovery hierarchy, circular economy.

THINKING IT THROUGH, STEP BY STEP

1. **Start from the forward chain.**

   Normal (forward) logistics moves new products from supplier → manufacturer → distributor → retailer → consumer. But products also **come back**: wrong size, broken, unsold, recalled, or at the end of their life.

2. **Definition.**

   Reverse logistics is the process of planning, implementing and controlling the efficient, cost-effective flow of products, materials and related information **from the point of consumption back to the point of origin**, in order to **recapture value or ensure proper disposal** (Rogers and Tibben-Lembke).

3. **What flows back, and why?**

   - **Customer returns**: wrong size, changed mind (very high in e-commerce), damaged in delivery.
   - **Product recalls**: safety or quality defects.
   - **Unsold stock**: seasonal goods, expired items, overstock.
   - **Repairs and warranty claims.**
   - **End-of-life products**: old phones, batteries, electronics (e-waste).
   - **Reusable packaging**: pallets, crates, returnable bottles and gallons.

4. **Why do firms bother? (drivers)**

   - **Economic**: recover value from parts and materials; resell refurbished items.
   - **Legal**: take-back and **extended producer responsibility (EPR)** laws, e-waste rules.

- **Customer service**: easy returns are an order winner online.
- **Environmental (ESG)**: less landfill, supports 3R and the circular economy (Q13).

5 **How does it work? The process, step by step.**

1. **Gatekeeping**: decide which returns to accept at the entry point (screening out invalid or fraudulent returns saves cost).
2. **Collection**: pick up from customers, stores or drop points.
3. **Sorting and inspection**: test and grade the item's condition.
4. **Disposition**: choose the best option for each item using the **recovery hierarchy** (most value kept first):
   - **Resource reduction** (design so less comes back)
   - **Reuse / resell** as is (return to stock, outlet, second-hand market)
   - **Remanufacturing / refurbishing** (repair and restore to "like new")
   - **Recycling** (recover materials)
   - **Disposal with energy recovery** (burn for energy)
   - **Landfill** (last resort)
5. **Redistribution**: send recovered goods or materials back into the forward chain.

6 **Why is it harder than forward logistics?**

- **Uncertain** timing, quantity and quality (you don't know what will come back, or when).
- Many **small, scattered** pickups, so it is costly.
- Each item needs **individual inspection** and a different route.
- Products often lose value quickly while waiting (fashion, electronics).
- Hence firms often outsource it to a **3PL** or use centralised return centres.

7 **Examples.**

- **AQUA 19-litre gallons** in Indonesia: empty gallons are collected when new ones are delivered, then cleaned and refilled (reuse).
- **Apple Trade In**: old iPhones are refurbished and resold, or disassembled (Apple's "Daisy" robot) to recover materials (recycling).
- **E-commerce returns** (Shopee, Tokopedia, Zalora): returned items are inspected and restocked or sent to outlets.

- **Product recalls** (e.g. car airbag recalls): defective units are brought back to be fixed.
- **H&M garment collecting**: used clothes collected in store for reuse or recycling.
- **Pallets and crates** returned and reused between partners.

8 **So what? Link to the big picture.**

Reverse logistics closes the loop: forward chain + reverse chain = a **closed-loop supply chain**, the operational heart of the **circular economy**. It turns returns from a cost into a source of value and supports the "planet" part of 3P and ESG.

ILLUSTRATION

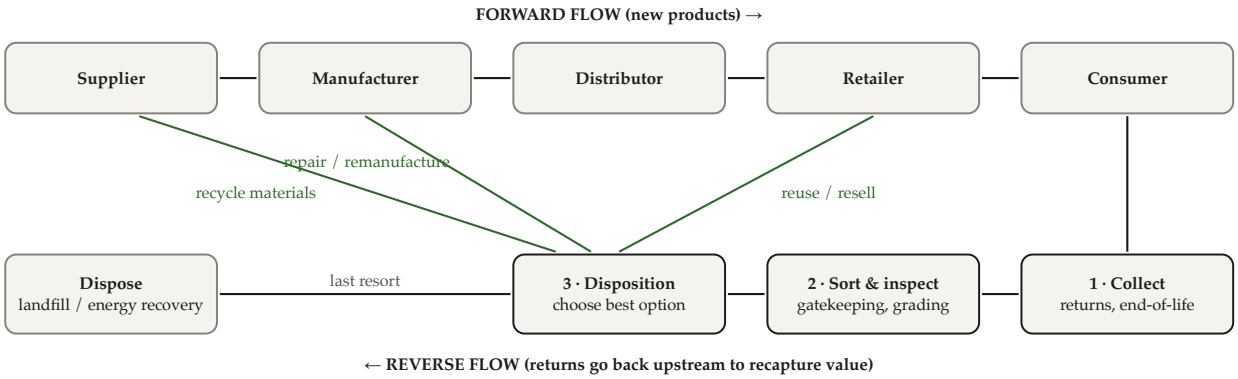


FIGURE 14.1 *Top: the forward flow. Bottom: the reverse flow: collect → sort and inspect → disposition, with each recovered item re-entering the chain at the highest possible level.*

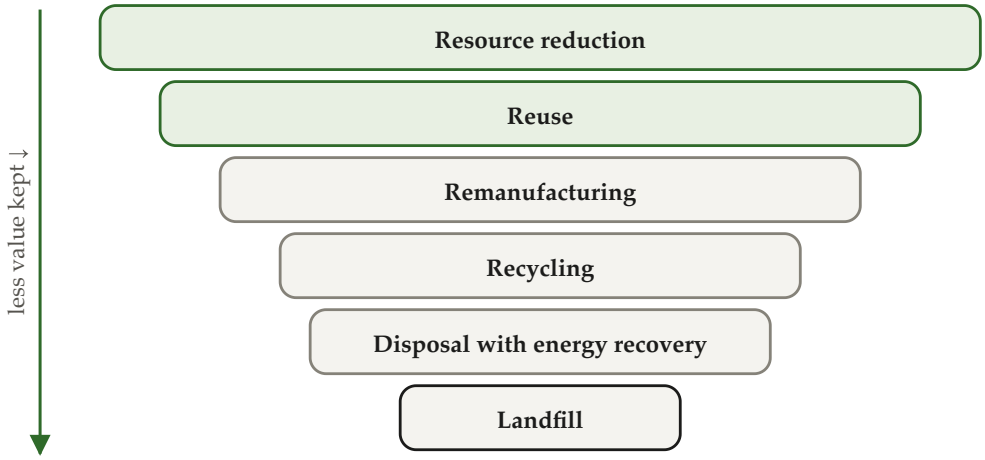


FIGURE 14.2 *The disposition step chooses the highest option on this ladder.*

Draw this in the exam: the forward chain as a row of boxes with arrows to the right; underneath, a second row with arrows to the left: Collect → Sort/Inspect → Disposition, with lines going back to the supplier (recycle), manufacturer (remanufacture) and retailer (resell). Add the funnel of recovery options beside it.

EXAM-READY ANSWER

MODEL ANSWER

Reverse logistics is the planning, implementation and control of the efficient flow of products, materials and information **from the point of consumption back to the point of origin**, to **recapture value or dispose of them properly**. It handles customer returns, product recalls, unsold or expired stock, repairs and warranty claims, end-of-life products and reusable packaging.

Firms do it for **economic** reasons (recovering value), **legal** reasons (take-back and extended producer responsibility laws), **customer service** (easy returns) and **environmental** reasons (supporting 3R, ESG and the circular economy).

The process is: **gatekeeping** (screen which returns to accept), **collection**, **sorting and inspection**, and **disposition**, where each item takes the highest option on the **recovery hierarchy**: resource reduction, reuse or resale, remanufacturing or refurbishing, recycling, disposal with energy recovery and, last, landfill. Recovered goods are then redistributed into the forward chain.

Reverse logistics is harder than forward logistics because timing, quantity and condition are uncertain, pickups are small and scattered, and each item must be inspected individually, so firms often use 3PLs or central return centres.

Examples: AQUA's returnable 19-litre gallons are collected, cleaned and refilled; Apple's Trade In refurbishes old iPhones or recycles their materials; e-commerce returns are inspected and restocked. Together with the forward chain, reverse logistics creates a **closed-loop supply chain**.

COMMON MISTAKES

✗ Defining it only as "returning goods". Add the purpose: **recapture value or proper disposal**.

✗ Skipping the process steps (gatekeeping → collection → sorting → disposition).

✗ Forgetting the recovery hierarchy, which the lecturer specifically pointed to.

APPENDIX · SUMMARY · KISI-KISI

One-page summary map (how the 14 questions connect)

THEME	QUESTIONS	ONE-LINE LINK
What we manage	Q3, Q4 (OM) → Q6 (SCM) → Q7 (logistics)	OM transforms inputs inside one firm; SCM coordinates many firms; logistics moves the goods between them.
How we compete	Q8 (winners/qualifiers), Q9 (network design), Q11 (vertical integration)	Know what wins the order, then design the network and ownership to deliver it.
How we run it well	Q10 (lean), Q5 (bullwhip), Q2 (forecasting), Q1 (LP)	Remove waste, share information, forecast well, and optimise decisions with data.
Where it came from	Q12 (industrial revolutions)	Each new technology raised productivity and widened supply chains.
What it must also protect	Q13 (ESG, 3P, CSR, 3R), Q14 (reverse logistics)	A modern chain is judged by people and planet too, and closes the loop.

Final exam tip: for any question, write the definition first, draw the diagram second, give one real example third, and finish with one sentence of evaluation ("this is better when…", "the risk is…"). That pattern covers what the marking scheme looks for.

Flashcards

Cover the right-hand column and answer from memory.

S1–2 Intro 8 cards

Define logistics (the 7 Rights)	Getting the right product, in the right way, in the right quantity and right quality, in the right place, at the right time, for the right customer, at the right cost.
Define supply chain management (SCM) and its 3 purposes	The management, across and within a network of upstream and downstream organisations, of relationships and flows of material, information and resources. Purposes: create value, enhance efficiency, satisfy customers.
The 3 key flows in SCM	(1) Physical flows of materials; (2) flows of data and information; (3) resources, especially finance but also people and equipment. Good inter-company relationships are an intangible resource.
Farm to fork / sketch to store / dust to rust / teeth to tail	Supply chain flows in food / clothing / cars and machinery / military.
Supply chain vs value chain	The value chain (entities, activities and competencies that add value) creates the flows; the supply chain (network of organisations and relationships) operationalises those flows.
6 developments behind logistics and SCM	Reduced transport intensity of freight, falling product prices, deregulation of transport, productivity improvements, emphasis on inventory reduction, changes in company structure.
Foundational premises FP1–FP3	FP1: the chain is a network of nodes and links. FP2: it is a complex adaptive system (tension between control and emergence). FP3: it is relative to a particular product and agent.
Foundational premises FP4–FP6	FP4: physical chain + support chain. FP5: bounded by the focal agent's visible horizon. FP6: that horizon attenuates with physical distance, cultural distance and closeness centrality.

S3–4 Lean 8 cards

Evolution of production strategies	Craft (custom, low volume) → mass production (standardisation, economies of scale) → flexible manufacturing → lean (waste elimination, continuous improvement) → digital: automation, Industry 4.0, sustainability.
4 Vs of process design	Volume, Variety, Variation in demand, Visibility of the customer.
The 7 wastes (muda) + the 8th	Overproduction, waiting, transportation, inappropriate processing, unnecessary inventory, unnecessary motion, defects; 8th: underutilisation of resources.
5 lean principles	Identify value (customer's view); map the value stream; make steps flow and eliminate waiting; pull, not push; find root causes and strive for perfection.
Lean: what shifts?	From push to pull; from make-to-stock (MTS) to make-to-order (MTO); level production plus just-in-time (JIT) inventory.
Lean consumption (6 rules)	Solve the customer's problem completely; don't waste their time; provide exactly what they want; where they want it; when they want it; keep aggregating solutions to cut their time and hassle.
Servitisation	Selling services combined with products. Anything-as-a-service (XaaS), e.g. mobility as a service (MaaS), supply chain as a service (SCaaS).
3 quality management approaches in S3–4	Total quality management (TQM), kaizen (continuous improvement), Lean Six Sigma.

S5–6 Systems 7 cards

Business process management (BPM): definition and phases	Systematic approach to model, analyse, redesign and monitor business processes. Phases: modelling, analysis, redesign, execution, monitoring.

Management information systems (MIS) in SCM	Computerised systems that collect and process data for decisions: enterprise resource planning (ERP), SCM execution systems, supplier portals. They give real-time visibility and exception management.
AirSupply case: features and benefits	Aerospace digital platform: demand forecasting, order tracking, vendor-managed inventory (VMI), early warning. Benefits: visibility, collaboration, reduced inventory and cost.
3 types of business analytics	Descriptive (monitor KPIs, key performance indicators), predictive (demand forecasting), prescriptive (recommend actions such as replenishment).
Simulation vs optimisation	Simulation (discrete-event, agent-based) tests what-if scenarios without disruption. Optimisation (linear/integer programming) finds the best design, inventory or schedule under constraints.
4 challenges of SCM systems and models	Data quality and integration; complexity of modelling dynamic global chains; resistance to change in process redesign; balancing robustness and agility.
Problem-solving methods (S5–6)	Root cause analysis (5 Whys, fishbone), Six Sigma DMAIC (Define, Measure, Analyse, Improve, Control), PDCA cycle (Plan, Do, Check, Act).

S7–8 Strategy 8 cards

Define outsourcing and a virtual organisation	Outsourcing: transfer to a third party of the management and delivery of a process previously done in-house. Virtual organisation: outsources most, if not all, major functions.
5 contents of a service level agreement (SLA)	Roll-out and duration; scope of services; areas of responsibility and legal requirements; performance metrics; termination/exit arrangements.
4 outsourcer–supplier relationship stages	Master-servant → consultative → peer-to-peer → competitive.

Outsourcing vs offshoring	Outsourcing = moving work to another company (who). Offshoring = moving work to another country, usually for lower cost (where).
Supplier tiers vs modules	Tiers: hierarchy of supplier levels. Modules: complete sub-assemblies from a lower-tier supplier to a higher tier or final customer.
Postponement	Delaying final production, customisation or distribution until demand is clearer / closer to the point of sale. Reduces inventory risk, increases flexibility, enables mass customisation.
Leagile supply chain	Lean (efficiency, waste reduction) for stable early-stage production + agile (flexibility, quick response) for the dynamic final stage, separated by the decoupling point.
Responsiveness vs agility	Responsiveness: meet customer demand quickly. Agility: adapt to market changes and disruptions.

S9–10 Sustainability 5 cards

Scope 1, 2, 3 emissions	Scope 1: direct (e.g. company trucks). Scope 2: indirect from generating the energy the company uses. Scope 3: all other indirect emissions upstream and downstream; hardest to measure, often the largest.
Carbon footprint formula	Fuel used × emission factor for that fuel, in kg CO_2eq (carbon dioxide equivalent, which includes methane and nitrous oxide).
Corporate social responsibility (CSR) vs environmental, social and governance (ESG)	CSR is the voluntary ethical foundation and vision; ESG gives the measurable targets and accountability to implement CSR.
Reverse logistics and the 3Rs	Material flows from consumption back toward origin for return, reuse, repair, remanufacture or recycling. Circular economy 3Rs: reduce, reuse, recycle.
4 ways to make logistics more sustainable	Redesign supply chains; use less harmful energy; use scale (larger single loads); promote efficiency solutions.

S11–12 Risk 7 cards

Vulnerability / robustness / resilience	Vulnerability: likelihood of being exposed to damage, disruption or failure. Robustness: manage regular demand fluctuations. Resilience: return to the original or desired state after disturbance.
Bullwhip effect and its 4 causes	Demand variability is amplified as orders move upstream. Causes: demand forecast updating, order batching, price fluctuations, rationing and shortage gaming.
8 types of supply chain complexity	Network, process, range, product, customer, supplier, organisational, information.
Supply chain risk management (SCRM) strategies	Visibility via information sharing; flexible production; postponement; multiple sourcing and flexible contracts; redundancy (safety stock, e.g. via VMI); joint planning.
4 types of supply chain risk	Systemic, market, operational, credit and liquidity.
VUCA and the response	Volatility → Vision, Uncertainty → Understanding, Complexity → Clarity, Ambiguity → Agility.
Transport security initiatives	IMO's ISPS code (International Maritime Organization, International Ship and Port Facility Security); US C-TPAT (Customs-Trade Partnership Against Terrorism), CSI (Container Security Initiative), ACAS (Air Cargo Advance Screening); EU AEO (Authorised Economic Operator); ISO 28000.

S13–14 Purchasing 5 cards

Define international purchasing	Obtaining goods or services from a market that requires crossing international boundaries; draws on logistics, marketing, negotiation, language, culture, technical, audit, standards and trade-finance skills.
The 8 steps of international purchasing	Identify requirements → supplier research & selection → negotiation planning → contract activation → manage logistics → cargo tracking → goods receipt & evaluation → after-care & continuous improvement.

3 international payment methods	Letter of credit (L/C: buyer's bank guarantees payment), documentary collection (export documents as payment security), open account (pay after receipt; highest risk for the supplier).
Free trade zone (FTZ) / bonded warehouse	Import duty is deferred until goods are withdrawn; goods can be withdrawn partially as needed.
3PL vs 4PL	Third-party logistics (3PL) performs logistics operations; fourth-party logistics (4PL) integrates and manages the whole chain, often coordinating several 3PLs.

Strategy toolbox 8 cards

Productivity formula	Output ÷ input. Multifactor productivity = output ÷ (labour + materials + overhead).
Order qualifier vs order winner	Qualifier: needed for a product to be considered at all. Winner: the criterion that makes the customer choose you over competitors.
4 competitive priorities and class examples	Cost (lean), speed (Zara 9-day design-to-rack), quality (Ritz-Carlton), flexibility (National Bicycle mass customisation).
VRIO (resource-based view)	Valuable, Rare, Inimitable, Organised to capture value (Jay Barney).
Fisher matrix	Functional products (stable demand) → efficient supply chain. Innovative products (unpredictable demand) → responsive supply chain.
Balanced scorecard (BSC) perspectives	Financial, customer, internal processes, learning and growth (Kaplan and Norton).
Porter's Five Forces	Rivalry among competitors, threat of new entrants, threat of substitutes, bargaining power of suppliers, bargaining power of buyers.
Little's Law	$L = \lambda W$: average number in the system = arrival rate × average time in the system.

Improvement 3 cards

4 approaches to improvement	Business process reengineering (BPR: radical), total quality management (TQM), lean, Six Sigma.
Kaizen vs innovation	Kaizen: small, continuous steps, everyone, low investment, people focus. Innovation: large, abrupt steps, few champions, large investment, technology focus.
Garvin's 8 dimensions of quality	Performance, features, reliability, conformance, durability, serviceability, aesthetics, perceived quality.

Forecasting 5 cards

Least-squares slope (b) and intercept (a)	$b = (\Sigma xy - n\cdot\bar{x}\cdot\bar{y}) \div (\Sigma x^2 - n\cdot\bar{x}^2)$; $a = \bar{y} - b\cdot\bar{x}$; forecast $\hat{Y} = a + bX$.
Multiple regression: Significance F and p-values	Significance $F < 0.05 \rightarrow$ the overall model is significant. A predictor's p-value $< 0.05 \rightarrow$ that variable is significant; otherwise consider dropping it.
Seasonal index	Average demand for that season ÷ overall average demand. The indices sum to the number of seasons.
MAD, MSE, MAPE	Mean absolute deviation = average $\lvert\text{error}\rvert$; mean squared error = average error^2; mean absolute percentage error = average $\lvert\text{error}\rvert \div$ actual $\times 100\%$. Lower is better.
Simple vs weighted moving average	A simple MA weights the last n periods equally; a WMA gives chosen weights (summing to 1), usually more to recent periods, so it reacts faster.

Self-check quiz

Pick an answer for each question, then check yourself against the answers and explanations that follow.

1. **Which is NOT one of the three key flows in supply chain management (SCM)?**
   - A. Physical materials
   - B. Data and information
   - C. Resources such as finance and people
   - D. Advertising campaigns

2. **'Sketch to store' describes the supply chain of…**
   - A. Food
   - B. Clothing
   - C. Cars and machinery
   - D. Military

3. **Foundational premise FP5 says the supply chain is…**
   - A. A network of nodes and links
   - B. Relative to a product and agent
   - C. Bounded by the focal agent's visible horizon
   - D. A physical chain plus a support chain

4. **According to S1–2, the value chain ___ the flows while the supply chain ___ them.**
   - A. operationalises / creates
   - B. creates / operationalises
   - C. measures / finances
   - D. outsources / controls

5. **Lean production shifts a firm from…**
   - A. Pull to push
   - B. Make-to-order (MTO) to make-to-stock (MTS)
   - C. Push to pull and MTS to MTO
   - D. Level production to batch production

6. **Excess finished goods built 'just in case' is which waste?**
   A. Waiting
   B. Overproduction
   C. Motion
   D. Defects

7. **Which of the 4 Vs of process design refers to how much the customer sees of the process?**
   A. Volume
   B. Variety
   C. Variation
   D. Visibility

8. **Recommending how much stock to reorder next week is which type of analytics?**
   A. Descriptive
   B. Diagnostic
   C. Predictive
   D. Prescriptive

9. **In the AirSupply case, the supplier monitors and refills the customer's stock. This is called…**
   A. Enterprise resource planning (ERP)
   B. Vendor-managed inventory (VMI)
   C. Business process management (BPM)
   D. Cross-docking

10. **A firm keeps undyed sweaters and dyes them only when colour demand is known. This is…**
    A. Offshoring
    B. Postponement
    C. Order batching
    D. Servitisation

11. In a leagile supply chain, lean principles are applied…

A. Downstream of the decoupling point
B. Upstream of the decoupling point, in stable early-stage production
C. Only in retail
D. Only in transport

12. Buyer and supplier interact as equals, with joint decisions and shared risk and reward. Which stage?

A. Master-servant
B. Consultative
C. Peer-to-peer
D. Competitive

13. Which is NOT listed as part of a service level agreement (SLA)?

A. Scope of services
B. Performance metrics
C. Termination/exit arrangements
D. The supplier's marketing plan

14. A trucking company's diesel emissions are, for that company…

A. Scope 1
B. Scope 2
C. Scope 3
D. Not reported

15. Electricity bought to power a warehouse is…

A. Scope 1
B. Scope 2
C. Scope 3
D. Carbon neutral

16. Which statement about corporate social responsibility (CSR) and environmental, social and governance (ESG) criteria is correct?

A. They are identical
B. CSR sets the ethical vision; ESG provides measurable targets
C. ESG is voluntary; CSR is law
D. CSR covers only governance

17. The ability of a supply chain to return to its original or desired state after a disturbance is…

A. Robustness
B. Vulnerability
C. Resilience
D. Complexity

18. Retailer orders swing more than consumer sales, and factory orders swing even more. This is the…

A. Bullwhip effect
B. Postponement principle
C. Kaizen effect
D. Leagile effect

19. 'Too wide a range of products offered' is which type of complexity?

A. Product
B. Range
C. Customer
D. Network

20. Which payment method carries the highest risk for the supplier (exporter)?

A. Letter of credit (L/C)
B. Documentary collection
C. Open account
D. All equal

21. Storing imported goods so that duty is paid only when they are withdrawn uses a…

A. Cross-dock
B. Free trade zone (FTZ) or bonded warehouse
C. Hub-and-spoke network
D. Inland container depot only

22. Innovative products with unpredictable demand should use (Fisher)…

A. An efficient supply chain
B. A responsive supply chain
C. A push system
D. A virtual organisation

23. A bank has 30 customers arriving per hour and each stays 0.2 hours. Average customers inside (Little's Law)?

A. 6
B. 15
C. 30
D. 150

24. Regression gives a = 19.99 and b = 1.77. The forecast for X = 9 is…

A. 21.76
B. 35.92
C. 37.69
D. 179.91

25. Forecast errors of +10, −10 and +4 give a mean absolute deviation (MAD) of…

A. 1.33
B. 8.00
C. 72.00
D. 24.00

26. Q3 average demand is 160 and the overall quarterly average is 120. The Q3 seasonal index is…

A. 0.75
B. 1.20
C. 1.33
D. 40

27. In multiple regression output, a predictor with p-value 0.31 is…

A. Highly significant
B. Not significant at the 5% level
C. The intercept
D. The R^2

Answers and explanations

1. **D.** The slides list materials, information and resources.
2. **B.** Fashion moves from a design sketch to the retail store.
3. **C.** FP5 = visible horizon. FP1 = network, FP3 = relative, FP4 = physical + support.
4. **B.** Value chain creates the flows; supply chain operationalises them.
5. **C.** Lean moves from push to pull and from make-to-stock to make-to-order.
6. **B.** Producing more or earlier than needed is overproduction.
7. **D.** Visibility = customer exposure to the process.
8. **D.** Prescriptive analytics recommends actions such as replenishment.
9. **B.** VMI = vendor-managed inventory.
10. **B.** Delaying final customisation downstream is postponement.
11. **B.** Lean for the stable early stage, agile for the dynamic final stage.
12. **C.** Peer-to-peer = partnership of equals.
13. **D.** SLA: roll-out and duration, scope, responsibilities/legal, metrics, termination/exit.
14. **A.** Direct emissions from owned vehicles are Scope 1.
15. **B.** Indirect emissions from purchased energy are Scope 2.
16. **B.** CSR = foundation and vision; ESG = measurable targets and accountability.
17. **C.** Resilience = recovery. Robustness = managing regular demand fluctuations.
18. **A.** Variability amplified upstream is the bullwhip effect.
19. **B.** Range complexity = too many products; product complexity = too many unique components.
20. **C.** Open account: goods are sent first and payment comes later, with no bank guarantee.
21. **B.** FTZs and bonded warehouses defer import duty.
22. **B.** Innovative → responsive; functional → efficient.
23. **A.** $L = \lambda W = 30 \times 0.2 = 6$.
24. **B.** $19.99 + 1.77 \times 9 = 35.92$.
25. **B.** $(10 + 10 + 4) \div 3 = 8$. Absolute values stop the signs cancelling.
26. **C.** $160 \div 120 = 1.33$.
27. **B.** $p > 0.05$ means not significant; consider removing it and rerunning.

Mock exam

A full practice paper in three sections: A = concept essays, B = case analysis, C = Forum 11 calculations. Each question names its Course Intended Learning Outcome (CILO), the course goal it tests. Write your answer on paper first, then turn to the model answers and score yourself with the marking points. Marks show how much detail is expected. The model answers use the slide definitions word for word wherever possible, because examiners usually mark against them.

Section A · Concepts (CILO 1)

Q1. Explain the difference between logistics and supply chain management (SCM). Describe the three key flows in SCM and give one example of each. [15 marks]

Q2. Explain lean production. List the five lean principles and the seven wastes (muda), and state the eighth waste added in recent years. [15 marks]

Q3. What is business analytics in supply chain management? Explain descriptive, predictive and prescriptive analytics with a supply chain example of each, and name two challenges of applying systems and models in SCM. [15 marks]

Q4. Explain the four stages of outsourcer–supplier relationship development and the contents of a service level agreement (SLA). [15 marks]

Section A · Concepts (CILO 2)

Q5. Define supply chain vulnerability, robustness and resilience. Explain the bullwhip effect and suggest three supply chain risk management (SCRM) measures. [15 marks]

Q6. Explain Scope 1, 2 and 3 emissions with logistics examples. How is the carbon footprint of fuel calculated, and how can logistics systems be made more sustainable? [15 marks]

Section B · Case analysis

Q7. Case: 'BatikNow', a Jakarta fashion brand, sells shirts in 30 colours and 5 sizes. It produces all variants in Central Java three months ahead. Each season, popular colours sell out while others are discounted heavily. Using Fisher's matrix, postponement and the leagile concept, recommend a supply chain strategy. [20 marks]

Q8. Case: 'SinarMotor' imports engine parts from one supplier in Germany. Payment is by open account. Last year a strike at the German port stopped supply for six weeks and a production line closed. As a supply chain consultant: (a) identify the risks; (b) recommend purchasing, payment and risk management improvements. [20 marks]

Section C · Calculation (Forum 11)

Q9. Monthly demand for months 1–6 is 100, 110, 105, 115, 120, 118. (a) Compute 3-month moving average forecasts for months 4–7. (b) Compute the mean absolute deviation (MAD), mean squared error (MSE) and mean absolute percentage error (MAPE) for months 4–6. (c) Compute a weighted moving average (WMA) for month 7 with weights 0.5, 0.3, 0.2 (most recent first). (d) Interpret. [20 marks]

Q10. X (period) = 1, 2, 3, 4, 5 and Y (sales) = 3, 5, 7, 8, 12. Use least squares to find the regression line and forecast Y for period 6. Explain what a and b mean. [15 marks]

Q11. Quarterly demand: Year 1 = 80, 120, 150, 90; Year 2 = 100, 140, 170, 110. (a) Compute the seasonal index for each quarter. (b) If total demand next year is forecast at 520, forecast each quarter. [15 marks]

Section C · Interpretation (Forum 11)

Q12. A multiple regression of weekly sales (Y) on advertising spend (X_1) and price (X_2) gives: $R^2 = 0.86$, adjusted $R^2 = 0.82$, Significance F = 0.001, coefficient $X_1 = 4.2$ ($p = 0.004$), coefficient $X_2 = -1.5$ ($p = 0.27$). Interpret the output and advise management. [15 marks]

Model answers

Q1

Logistics is getting the right product, in the right way, quantity and quality, to the right place at the right time, for the right customer at the right cost (the 7 Rights). SCM is broader: it manages, across and within a network of upstream and downstream organisations, both the relationships and the flows of material, information and resources, with the purposes of creating value, enhancing efficiency and satisfying customers. Logistics is therefore a key component of SCM, not the same thing. The three flows are: (1) physical materials, e.g. bottles moving from a bottling plant to a distributor; (2) data and information, e.g. a retailer's point-of-sale (POS) data and purchase orders sent upstream; (3) resources, especially finance (payments to suppliers) but also people and equipment. Good inter-company relationships are an intangible resource that the slides call highly important.

Marking points

- 7 Rights definition of logistics (3)
- SCM definition with relationships + flows + network (4)
- SCM purposes: value, efficiency, customer satisfaction (2)
- Logistics is a component of SCM (2)
- Three flows each with a relevant example (4)

Q2

Lean production eliminates waste in a pull-based value stream: it shifts from push to pull, from make-to-stock (MTS) to make-to-order (MTO), and uses level production with just-in-time (JIT) inventory. The five principles are: identify value from the customer's perspective; map the value stream; make all steps flow efficiently and eliminate waiting; pull, not push (produce in response to customer demand); and identify root causes of problems and strive for perfection. The seven wastes are overproduction, waiting, transportation, inappropriate processing, unnecessary inventory, unnecessary motion and defects. The eighth is the underutilisation of resources (including people's skills).

Marking points

- Definition: push→pull, MTS→MTO, level production, JIT (4)
- Five principles in order (5)
- Seven wastes (5)

- Eighth waste (1)

Q3

Business analytics uses data-driven techniques (statistical analysis, predictive modelling, data mining) to extract insights from supply chain data, so firms can optimise processes, decide better, cut costs and become more responsive. Descriptive analytics monitors what happened, e.g. key performance indicator (KPI) dashboards in a control tower showing on-time delivery. Predictive analytics estimates what will happen, e.g. forecasting next month's demand with regression or moving averages. Prescriptive analytics recommends what to do, e.g. suggesting replenishment quantities. Challenges include: data quality and integration (partners' systems don't communicate, creating silos); the complexity of modelling dynamic global chains; resistance to change in process redesign; and balancing robustness against agility.

Marking points

- Definition of business analytics (3)
- Descriptive + example (3)
- Predictive + example (3)
- Prescriptive + example (3)
- Any two challenges explained (3)

Q4

In the master-servant stage the relationship is transactional and hierarchical: the outsourcer dictates terms, communication is one-way and the focus is control and cost. In the consultative stage the outsourcer asks for the supplier's input to improve performance; communication is two-way but the outsourcer still leads. In the peer-to-peer stage both act as equals, with trust, transparency, joint decision-making, long-term commitment, and shared risk and reward. In the competitive stage several suppliers compete on value, service and innovation, and the outsourcer uses that competition to keep driving improvement. An SLA is the formal agreement on the standards the provider must meet. It covers the roll-out and duration of the service, the scope of services, areas of responsibility and legal requirements, performance metrics (e.g. on-time delivery rate, inventory accuracy, response speed), and termination or exit arrangements.

Marking points

- Each stage correctly described (4 × 2 = 8)
- SLA definition (2)
- Five SLA contents (5)

Q5

Vulnerability is the likelihood of a supply chain being exposed to damage, disruption or failure. Robustness is the ability to manage regular fluctuations in demand. Resilience is the ability to return to the original or desired state after being disturbed; it takes a whole-system view, accepts that disturbances happen, and implies adaptability. The bullwhip effect is the amplification of demand variability as orders move upstream from consumer to retailer, wholesaler and manufacturer. It is caused by demand forecast updating, order batching, price fluctuations and rationing/shortage gaming. SCRM measures include increasing visibility through information sharing (e.g. sharing POS data), flexible production, postponement, multiple sourcing with flexible vendor contracts, redundancy through safety stock (e.g. vendor-managed inventory, VMI), and joint planning with partners.

Marking points

- Three definitions (6)
- Bullwhip definition (3)
- Causes (2)
- Three SCRM measures (4)

Q6

Scope 1 covers a company's direct emissions, e.g. its own trucks burning diesel. Scope 2 covers indirect emissions from generating the energy the company uses, e.g. electricity for a warehouse. Scope 3 covers all other indirect emissions upstream and downstream, e.g. suppliers' production and outsourced carriers; it is hard to measure and often the largest share. The carbon footprint of fuel = fuel used × the emission factor for that fuel, in kg CO_2eq (carbon dioxide equivalent, which also captures methane and nitrous oxide); about 95% of greenhouse gas emissions are CO_2. Logistics becomes more sustainable by redesigning supply chains to remove unnecessary freight movements, using less harmful energy sources, using scale (larger single loads that cut unit cost and emissions per unit), promoting efficiency, developing fuel-efficient engines, and using reverse logistics and the 3Rs (reduce, reuse, recycle).

Marking points

- Scope 1, 2, 3 with examples (6)
- Formula + CO_2eq meaning (3)
- At least three improvement measures (6)

Q7

BatikNow's shirts are innovative, fashion products with unpredictable demand by colour, so by Fisher's matrix they need a responsive supply chain. Making every variant months ahead is an efficient (push, make-to-stock) chain, a mismatch that explains the stockouts and markdowns. I recommend a leagile chain. Upstream, keep lean: buy fabric in bulk, produce plain (undyed) shirts in standard sizes with level production and economies of scale, because demand for the generic shirt is stable. Place the decoupling point at a stock of undyed shirts close to the market, e.g. near Jakarta. Downstream, be agile: apply postponement by dyeing or printing colours only after real point-of-sale (POS) data or orders show which colours sell, then deliver quickly. Benefits: lower inventory risk, fewer markdowns, fewer stockouts and mass customisation. Requirements: flexible dyeing capacity, real-time information sharing with stores, and possibly a 3PL for fast distribution. Risks: higher unit cost in the agile stage, and dye capacity becoming a bottleneck in peak season.

Marking points

- Classify product: innovative → responsive (Fisher) (4)
- Diagnose mismatch: MTS push → stockouts and markdowns (3)
- Leagile design: lean upstream, decoupling point, agile downstream (5)
- Postponement explained and applied (4)
- Benefits and requirements/risks (4)

Q8

(a) The main risks are supplier concentration (single sourcing), long and distant lead times, exposure to external shocks such as strikes and transport disruption, low visibility, and foreign exchange (FX) risk on euro payments. Open account favours SinarMotor as the buyer, but it may strain the relationship or the German supplier's willingness to extend credit. The six-week stoppage shows high vulnerability and low resilience. (b) Purchasing: follow the eight-step international purchasing process with better supplier research, including a second qualified supplier (multisourcing),

possibly in another region (multishoring), and use a 3PL or 4PL (fourth-party logistics provider) with online cargo tracking. Risk management: hold strategic safety stock, for example in a bonded warehouse or free trade zone (FTZ) so duty is paid only when parts are withdrawn; agree flexible contracts; plan jointly with the supplier; use alternative transport modes and ports in parallel. Payment and finance: consider a letter of credit (L/C) for new suppliers, manage FX risk (e.g. hedging), and insure goods in transit. These steps raise robustness and resilience at an acceptable cost.

Marking points

- Risks identified, including single sourcing, external shock, FX (5)
- Purchasing process / multisourcing / 3PL-4PL (5)
- SCRM: safety stock, FTZ/bonded, flexible contracts, joint planning, parallel modes (6)
- Payment and finance (L/C, FX, insurance) correctly reasoned (4)

Q9

(a) F4 = (100 + 110 + 105) ÷ 3 = 105.00; F5 = (110 + 105 + 115) ÷ 3 = 110.00; F6 = (105 + 115 + 120) ÷ 3 = 113.33; F7 = (115 + 120 + 118) ÷ 3 = 117.67. (b) Errors (actual − forecast): 10, 10, 4.67. MAD = (10 + 10 + 4.67) ÷ 3 = 8.22. MSE = (100 + 100 + 21.78) ÷ 3 = 73.93. MAPE = (10/115 + 10/120 + 4.67/118) ÷ 3 = (8.70% + 8.33% + 3.95%) ÷ 3 = 6.99%. (c) WMA7 = 0.5 × 118 + 0.3 × 120 + 0.2 × 115 = 59 + 36 + 23 = 118.0. (d) All errors are positive, so the moving average under-forecasts a rising trend (it lags). On average the forecast misses by about 8.2 units, or 7%. The WMA reacts faster because it weights recent months more. If the trend continues, a regression (trend) method may give lower MAD/MSE/MAPE.

Marking points

- (a) four forecasts (4)
- (b) MAD (3), MSE (3), MAPE (3)
- (c) WMA (3)
- (d) interpretation: lag, meaning of errors, method choice (4)

Q10

n = 5; $\Sigma x = 15$, $\bar{x} = 3$; $\Sigma y = 35$, $\bar{y} = 7$; $\Sigma xy = 3 + 10 + 21 + 32 + 60 = 126$; $\Sigma x^2 = 1 + 4 + 9 + 16 + 25 = 55$. $b = (\Sigma xy - n\cdot\bar{x}\cdot\bar{y}) \div (\Sigma x^2 - n\cdot\bar{x}^2) = (126 - 105) \div (55 - 45) = 21 \div 10 = 2.1$. $a = \bar{y} - b\cdot\bar{x} = 7 - 2.1 \times 3 = 0.7$. Line: $\hat{Y} = 0.7 + 2.1X$. Forecast for period 6: 0.7 + 2.1 × 6

= 13.3. Meaning: b = 2.1 means sales increase by 2.1 units each period on average (the trend); a = 0.7 is the line's value at X = 0, its starting level. In Excel: =SLOPE, =INTERCEPT and =FORECAST.LINEAR give the same results.

Marking points

- Sums correct (4)
- b correct (3)
- a correct (3)
- Forecast 13.3 (2)
- Interpretation of a and b (3)

Q11

(a) Quarter averages: Q1 = (80 + 100) ÷ 2 = 90; Q2 = 130; Q3 = 160; Q4 = 100. Overall average per quarter = (90 + 130 + 160 + 100) ÷ 4 = 120. Indices: Q1 = 90 ÷ 120 = 0.750; Q2 = 1.083; Q3 = 1.333; Q4 = 0.833 (sum = 4.00). (b) Average per quarter next year = 520 ÷ 4 = 130. Forecasts: Q1 = 130 × 0.750 = 97.5; Q2 = 130 × 1.083 = 140.8; Q3 = 130 × 1.333 = 173.3; Q4 = 130 × 0.833 = 108.3 (total ≈ 520). Q3 is the peak season (33% above average) and Q1 the low season (25% below).

Marking points

- Quarter averages and overall average (4)
- Four indices (4)
- Quarterly forecasts (5)
- Interpretation (2)

Q12

$R^2 = 0.86$ means 86% of the variation in weekly sales is explained by advertising and price together; adjusted $R^2 = 0.82$ corrects for the number of predictors and is used to compare models. Significance F = 0.001 < 0.05, so the overall model is statistically significant. Advertising is significant (p = 0.004 < 0.05): holding price constant, each extra unit of advertising spend raises sales by 4.2 units on average. Price has a negative coefficient (−1.5), as expected, but it is not significant (p = 0.27 > 0.05), so in this data it is not a reliable predictor. Advice: drop price and rerun the regression (or collect more data with greater price variation), then use the model with planned advertising spend to forecast sales. Remember regression shows association, not guaranteed causation.

Marking points

- R^2 and adjusted R^2 (4)
- Significance F (3)
- Advertising coefficient and p-value (3)
- Price coefficient and p-value (3)
- Recommendation (2)